



Invited Review

A review of composition evolution in Ni-based single crystal superalloys

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ARTICLE INFO

Article history:

Received 28 June 2019

Received in revised form 1 November 2019

Accepted 5 November 2019

Available online 8 January 2020

Keywords:

Nickel-based single crystal superalloys

Composition

Creep properties

Refractory elements

ABSTRACT

Due to the outstanding creep performance, nickel-based single crystal superalloys (Ni-SXs) are extensively applied in modern aero-engine and industrial gas turbine. Apart from the special single crystal structure which is disadvantageous to extension of creep cracks, Ni-SXs derive the creep strength from intrinsic two-phase microstructure (γ phase and γ' phase). Main microstructural parameters including volume fraction of γ' phase and the lattice misfit, and the formation and distribution of precipitated phase are determined by the compositions of alloys. Besides, the creep properties are greatly influenced by these microstructural parameters and precipitated phase. This review has summarized the relationships between different alloying elements and microstructures and indicated their influence on creep properties of Ni-SXs. In addition, with the improvements of experimental methods and characterization technique, some recent discoveries have provided additional evidence to support or challenge the previous creep theories of superalloys. In view of these new discoveries, this review has provided some perspectives which can be referenced in future compositional design of Ni-SXs.

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1. Introduction

Aerial gas turbine engines always face to very severe and complex work conditions consisting of high temperature, high stress and highly oxidized and/or corrosive environment, which raise vigorous requirements of high performance superalloys [1–4]. Nickel-based single crystal superalloys (Ni-SXs) are extensively applied in turbine blades because of their excellent thermo-mechanical performance, especially the high temperature creep properties [5–7]. As one of the most important failure mechanisms, creep deformation commonly acts on alloys to reduce the high temperature strength and resulted service life. Compared to traditional equiaxed superalloys, the creep properties of Ni-SXs are greatly enhanced [8,9]. The main reason can be seen as the elimination of grain boundaries of single crystal alloys since grain boundaries are weak parts for nucleation and propagation of creep cracks [10–12]. In addition, Ni-SXs derive the creep strength from intrinsic two-phase structures (γ phase and γ' phase). Like pure Ni, the γ phase has FCC crystal structure and forms solid solution in alloys. The distribution of γ phase is continuous that it serves as matrix to contain other precipitated phase. As the primary precipitated phase, γ' phase generally accounts for a large proportion in Ni-SXs and plays the most important role in strengthening alloys [13–15]. The γ' phase exhibits the L1₂ crystal structure in the form of precipitate phase and forms coherent interfaces with continuous γ matrix, which demonstrates as narrow channels between the γ' cuboids (Fig. 1(a)) [16]. Such the highly ordered structure significantly enhances the strength and creep resistance of alloys deriving from the high volume fraction in alloy. Since it is not easy for dislocations to cut into the rigid γ' cuboids that the movements of dislocation are mainly limited in the γ channels.

The compositions of some representative Ni-SXs are listed in Table 1. Ni and other minor elements such as C, B and Hf are not included in this table [17–21]. Although a wide variety of alloying additions in these alloys were chosen, some basic elements were used commonly including significant amounts of Ni, Cr, Co, Al, Mo, W, Ti, Ta, Re and Ru [22]. According to different roles in strengthening alloy, these elements are divided into two categories including γ phase elements and γ' phase elements [23–28]. The γ phase elements denote alloying elements which mainly distribute in γ phase and form solid solutions, such as Co, Cr, Mo, W, Re and Ru [29,30]. The γ phase is strengthened by adding these elements which significantly increase the solidus temperature to enhance the thermal stability [31] and decrease the stacking fault energy (SFE) to enhance the creep properties [32]. Also the microstructures of

alloys are expected to be optimized by designing the composition of γ phase thus enhancing the creep properties. For example, the most discussed elements Re and Ru in new generation Ni-SXs are expected to stabilize the microstructures during of creep. For element Re, it was shown to co-segregate with other γ elements at the γ/γ' interfaces thus restrain the interfacial degeneration and resulted coarsening of γ' precipitates [33,34]. Element Ru strongly partitions to γ matrix, thus reversely decreasing the partitioning coefficient of other γ phase elements in the matrix such as Re and Cr [35,36]. Therefore the precipitation of topologically close-packed (TCP) phase is greatly restrained to enhance the creep properties of alloy. The γ' phase elements mainly include Al, Ti and Ta. In the formation of microstructures with these elements, the construction of Ni-X bonds, where X denotes Al, Ti or Ta, are promoted rather than Ni-Ni or X-X structures. These bonds demonstrate a strong degree of chemical order, giving rise to the formation of ordered γ' phase to make them different to disordered γ phase with FCC structures [37,38]. The additions of γ' phase elements need to be considered in two aspects. To adjust the volume fraction of γ' phase, it is needed to change the total contents of γ' phase elements. Besides, some alloying elements such as Ta and Ti are used to replace Al that improves the intrinsic strength of γ' phase. Fig. 1(b) gives the series of EDX maps of common alloying elements Al, Co, Cr, Ni, Re, Ta, Ti, and W within a Ni-SX, showing the standard partitioning behavior of these additions.

It is obvious that the microstructures of alloys could change with variations of alloying elements [39–42]. To establish the relationships between alloying elements and creep properties, we need to consider the effects of compositions on a variety of microstructural parameters of Ni-SXs, such as the difference of lattice constants between the γ and γ' phase, which determines the lattice misfit [43–47], the density and integrity of formed dislocation networks [48–51], and the distribution and composition of TCP phase [52–55]. The creep properties of alloy tightly depend on the interfacial coherency, which is generally described by the lattice misfit, δ , defined as $\delta = [(a_{\gamma'} - a_{\gamma}) / (a_{\gamma'} + a_{\gamma})]$. The parameter $a_{\gamma'}$ and a_{γ} denote the lattice constants of γ' phase and γ phase, respectively. Since the lattice constants greatly depend on the partitioning of alloying additions in the γ' and γ phase, the lattice misfit can be either positive or negative in sign according to the value of lattice constants of the two phases. On the one hand, solutes such as Ta and Ti, which preferably partition to the γ' phase have larger atomic size than ones in γ phase such as Co and Cr. The partitioning of these large solutes in γ' phase can replace some of the atomic sites of the basic Ni-Al structure, thus effectively increasing the lat-

Table 1
Compositions of some representative Ni-based single superalloys (wt%).

Generation	Alloy	Co	Cr	Mo	W	Al	Ti	Ta	Re	Ru
First	PWA1480	5.0	10.0	–	4.0	5	1.5	12.0	–	–
	Rene N4	8.0	9.0	2.0	6.0	3.7	4.2	4	–	–
	CMSX-2	4.6	8.0	0.6	8.0	5.6	1.0	9.0	–	–
	AM1	6.0	8.0	2.0	6.0	5.2	1.2	9.0	–	–
	SRR99	5.0	8.0	–	10.0	5.5	2.2	3.0	–	–
	TMS-6	–	9.2	–	8.7	5.3	–	10.4	–	–
Second	PWA1484	10.0	5.0	2.0	6.0	5.6	–	9.0	3.0	–
	Rene N5	8.0	7.0	2.0	5.0	6.2	–	7.0	3.0	–
	CMSX-4	9.0	6.5	0.6	6.0	5.6	1.0	6.5	3.0	–
	TMS-82	7.8	4.9	1.9	8.7	5.3	0.5	6.0	2.4	–
	Rene N6	12.5	4.2	1.4	6.0	5.75	–	7.2	5.4	–
Third	CMSX-10	3.0	2.0	0.4	5.0	5.7	0.2	8.0	6.0	–
	TMS-75	12.0	3.0	2.0	6.0	6.0	–	6.0	5.0	–
	PWA1497	16.5	2.0	2.0	6.0	5.55	–	8.25	5.95	3.0
Fourth	MC-NG	–	4.0	5.0	1.0	6.0	0.5	5.0	4.0	4.0
	TMS-138	5.8	3.2	2.9	5.9	5.8	–	5.6	5.0	2.0
Fifth	TMS-162	5.8	3.0	3.9	5.8	5.8	–	5.6	4.9	6.0
	TMS-196	5.6	4.6	2.4	5.0	5.6	–	5.6	6.4	5.0
Sixth	TMS-238	6.54	4.6	1.1	4.0	5.9	–	7.6	6.4	5.0

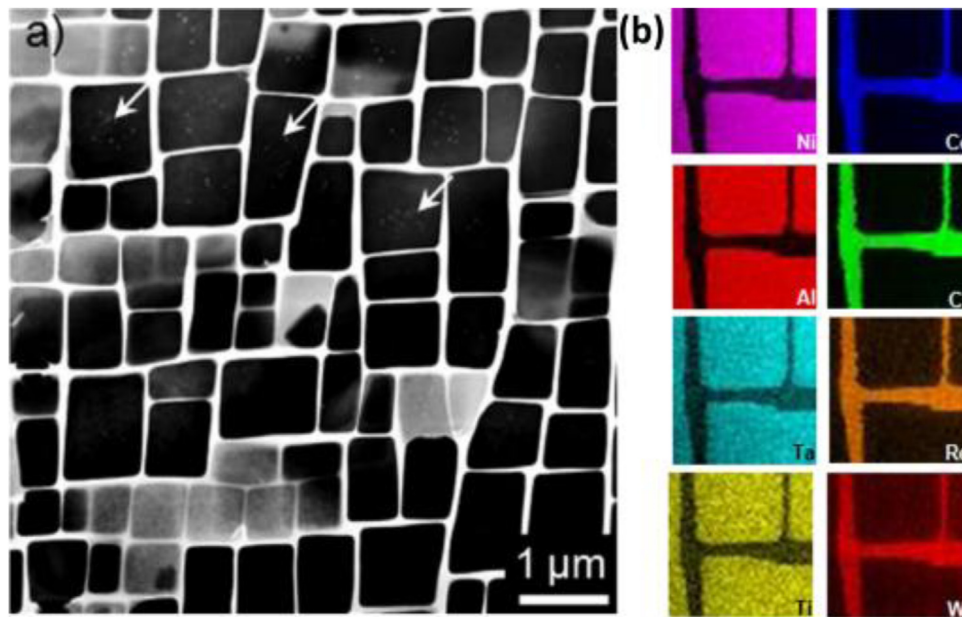


Fig. 1. Standard γ/γ' microstructure and elemental partitioning behavior of Ni-SXs. Adapted from Ref. [16]. (a) HAADF-STEM microscopy of a Ni-SX. Cuboidal γ' particles (dark) are separated by thin γ channels (bright); (b) Series of EDX maps showing alloying elements Ni, Al, Ta and Ti partition to the γ' phase and Co, Cr, Re and W partition to the γ matrix, respectively.

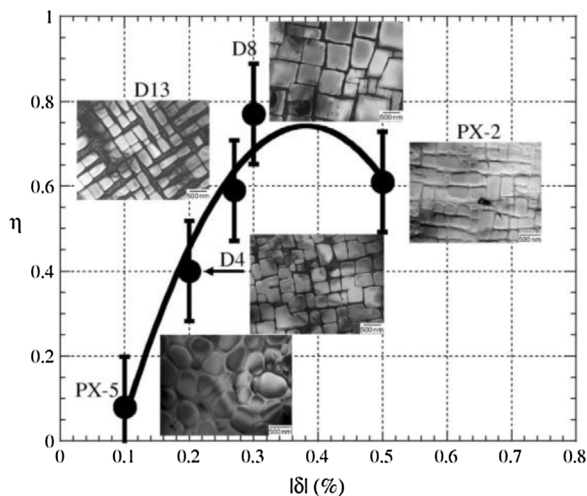


Fig. 2. Effects of lattice misfit on shape of γ' phase. Adapted from Ref. [65].

tice constant of γ' phase [56,57]. Even if the γ phase elements also have larger atomic size than pure Ni and additions of these solutes can increase the lattice constant of γ phase, the increasing degree can be expected to be lower than that in γ' phase. Therefore, most Ni-SXs have the positive lattice misfits at the room temperature [58,59]. However, the positive lattice misfits of Ni-SXs generally transferred to negative at high temperature due to the much larger thermal expansion coefficient of γ phase [60,61]. It was suggested the γ/γ' microstructure experienced significant variation with the changed lattice misfit because of the changed interfacial coherency stress, which acts as one of the main driving force for coarsening of the γ' precipitates [62]. Reduced lattice misfit implies the decrease of coherency strains and the decreased coarsening kinetics. Near zero lattice misfit can produce the closely spherical γ' precipitates in order to diminish the surface energy. As lattice misfit increased, the interfacial energy also increased to minimize the enhanced elastic strain energy and the large lattice distortion energy, thus leading to the cubical change of γ' precipitations, as shown in Fig. 2

[63]. If the lattice misfit is not too large, the γ/γ' interface remains its coherency with low interfacial energy to stabilize the interface. Also a cubical shape of γ' precipitate was shown to be beneficial to creep properties of alloy because of the minimized elastic moduli of both the two phases along $\langle 100 \rangle$ direction [64]. In alloy design, elements W and Mo are generally applied to enhance the lattice misfit in order to optimize the shape of γ' precipitates because of their much larger atomic size than Ni. A moderate lattice misfit is expected to be advantageous to creep properties, and generally an absolute value of lattice misfit around 0.4 is favorable to many Ni-SXs [65]. However, an overlarge lattice misfit would give rise to irregular coarsening of γ' particles because of the loss of interfacial coherency thus destabilizing the microstructures and reversely affecting the creep properties of alloy, as shown in Fig. 2 [65]. Therefore the additions of W and Mo are limited in minor contents for most Ni-SXs.

During creep, dislocations propagate at the γ/γ' interface to form dislocation networks, which help to restrain the movements of dislocations. Researches have shown that a dense dislocation network is advantageous to the creep properties of alloy because of the expected restrained effects on dislocation movements [64,66]. And additions of Mo [67–69] and Ru [70,71] were seen to be beneficial to forming the dense and regular dislocation networks. However, some recent researches have shown very dense dislocations aggregate to γ/γ' interface can form diffusive channels which promote the inter-diffusion of solutes across the interface and destabilize the γ/γ' microstructure [72–74]. To obtain high temperature performance, significant amounts of refractory elements such as Cr, Re, W and Mo are applied in Ni-SXs. However, excessive additions of these elements can accelerate the precipitation of TCP phase [75–78]. The formation of TCP phase greatly consumes γ phase and γ' phase elements leading to the dissolution of γ/γ' microstructure around it [79–81]. And the big size and irregular shape of TCP phase lead to great stress concentration around it which makes it easily for the nucleation and extension of micro-cracks [53,82]. Therefore the total amounts of these refractory elements in Ni-SXs need to be restricted. A tendency in design of new Ni-SXs is to enhance the amounts of Re and Mo with reducing the contents of W and Cr [83–85]. Additions of Ru began at the

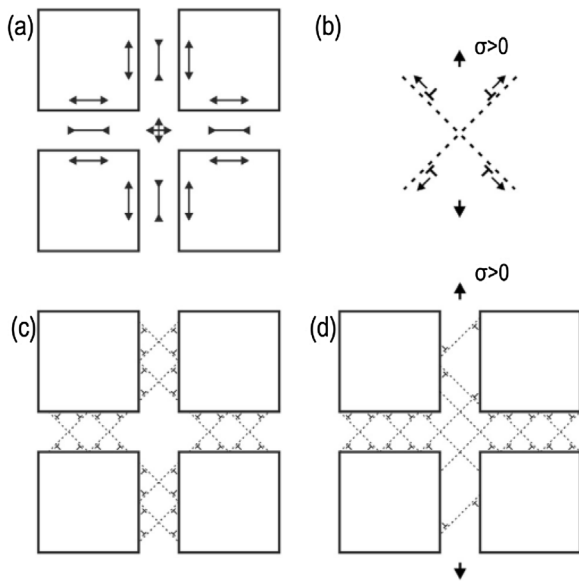


Fig. 3. Schematic illustrations of dislocation movements in the γ channels. Adapted from Ref. [72]. (a) Stress components associated with misfit; (b) Effect of an external tensile stress on two dislocation dipoles; (c) Misfit dislocations. (d) Interface dislocations after creep.

fourth generation Ni-SXs to enhance the creep properties of alloy, because it was considered to restrain the formation of TCP. But the internal mechanism is still unclear and needs to discuss more.

2. Creep mechanisms and microstructural evolution during creep of Ni-SXs

2.1. Dislocation motion in γ phase

After a full heat treatment very few dislocations exist in γ matrix and the γ' precipitates are essentially dislocation-free prior to creep [66]. Under external stress, some individual dislocation segments can act as dislocation sources giving rise to the multiplication of dislocations to fill in the γ matrix and form the dislocation networks around the γ' precipitates. The movement of dislocations requires the diffusion of atoms/vacancies from/to dislocations [86]. Generally, in a large temperature ranges, the dislocation movements occurring in the γ matrix cause the main creep deformation of Ni-SXs because of the great resistance of γ' precipitates to dislocation motion [87]. Apart from applied stress, the ingrown misfit stress serves also as an important driving force for dislocation motion. It was shown the lattice misfit stress greatly affects the driving force for dislocations to overcome the Orowan resistance and the evolution of γ/γ' interfacial dislocation networks [88]. With the movements of dislocations, the lattice misfit stress is gradually relieved. And it is difficult to quantitatively clarify the effects of lattice misfit stress on the dislocation motion with considering the effects of external stress [89]. However, the combined stress in the γ matrix can be very different that gives rise to the non-uniform distribution of stress around the γ' precipitates [90,91]. Simple rules were presented in previous researches to clarify the effects of lattice misfit on dislocation movements. The systems where the external stress and the misfit stress produce forces of the same sign are preferred for glide of dislocations in γ matrix [92]. In alloy with negative lattice misfit, a tensile stress drives dislocations in the γ channels parallel to the stress axis to normal channels to accommodate the misfit stress (Fig. 3). Thus the high density of dislocations in normal channels helps to form dense dislocation networks with small network spacing [93,94]. The different distribution of dislo-

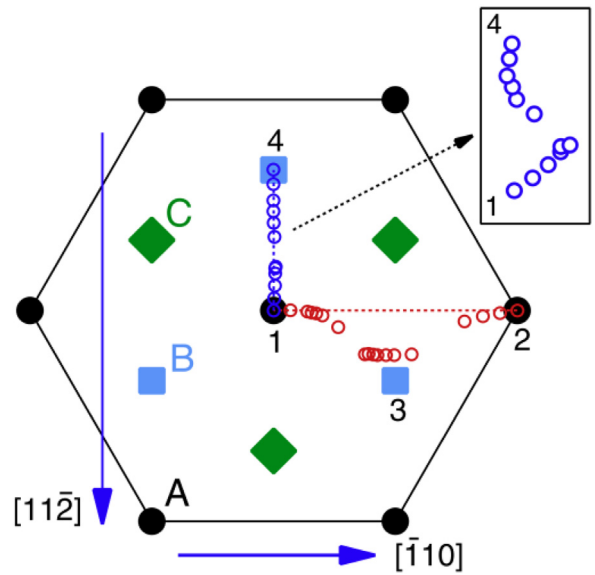


Fig. 4. Schematic illustrations of the formation of stacking faults referring to the dislocation glide in γ matrix. Adapted from Ref. [96]. Dislocation decomposition under $(1\ 11) \langle -111 \rangle$ alias shear deformation: $a/2[-1\ 10] \rightarrow a/6[-2\ 11] + a/6[-1\ 2\ -1]$ ($12 \rightarrow 13 + 32$).

cations between the vertical and horizontal channels can influence the dislocation motion. In alloys with negative misfits, the tensile stress promotes the movement of dislocations in the γ sheets normal to the stress axis, while the compressive stress promotes the movements in the sheets parallel to axis. In addition, by changing the sign of misfit, the dislocations move in a reversed pattern [66]. As result, under same tensile loading, positive lattice misfit produces a lower deformation rate than negative lattice misfit [95].

In a macroscopic sense, when a full dislocation $a/2 \langle 110 \rangle$ glides along the (111) plane, it is decomposed to two partial dislocations, which are separated by a distance. The glide of the first partial dislocation creates a stacking fault, which is removed by the passing of second one. As shown in Fig. 4, the reaction is in the form of $a/2[-1\ 10] \rightarrow a/6[-2\ 11] + a/6[-1\ 2\ -1]$ [96]. The partial dislocations are in mix form that the screw components are in opposite sign to provide an attractive force, while the edge ones point to the same direction to offer a repulsive force. The dissociation of partial dislocations drives the formation of intrinsic stacking faults, since the magnitude of the repulsive force is large than the attractive force. And the difference between the two forces is exhibited in the form of SFE. To hypothesize the movement of partial dislocations is restricted in a single (111) plane without other decomposition reactions, unless the partial dislocations recombine to a full screw dislocation, the deformation is restrained in the (111) plane without the required cross-slip pattern of screw dislocations. The meaning of SFE derives from the required energy for forming a stacking fault of unit area, which expresses the difficulty of forming an intrinsic stacking fault. In addition, SFE is relative to the width of stacking fault. A smaller SFE implies fewer energy to form the stacking fault, thus the width is larger. More importantly, a larger width represents more energy required for recombining the partial dislocations, implying the higher strength of γ matrix [97,98]. It is a sufficient way to strengthen the γ phase by reducing the SFE. Based on first-principles calculations, Shang et al. calculated the SFE of 26 kinds of two phase Ni-X dilute alloys indicating all alloying elements in this experiment can decrease the SFE of fcc-Ni [99]. The degrees of reduced SFE are in a sequence approximately according to the site of alloying elements X in the elemental table that the further X is from Ni, the larger decrease of SFE for the Ni-X alloys. Besides, the values of SFE for all Ni-X systems decrease with

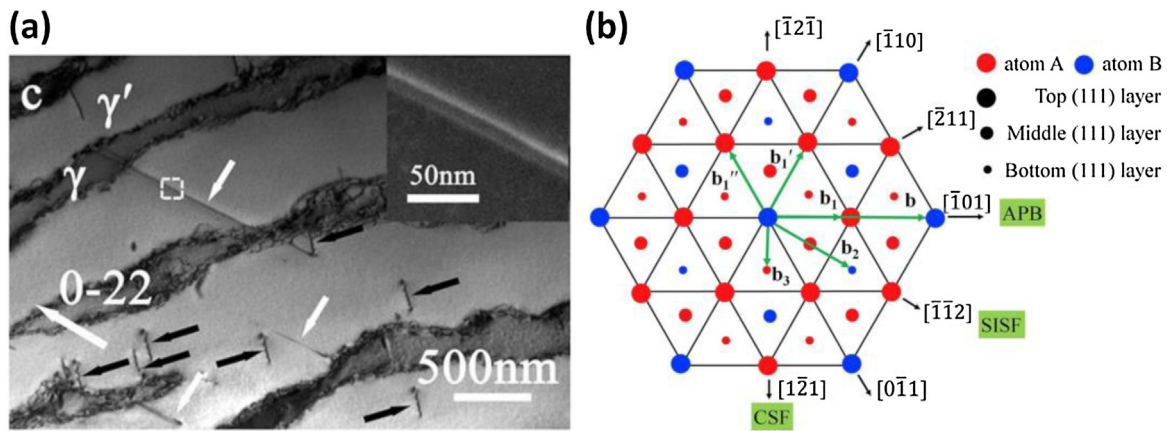


Fig. 5. Illustrations of dislocation motion in γ' phase with forming APBs. (a) Dislocations in γ matrix travel in pairs across the γ' phase; (b) Schematic illustration of the movements of dislocation pairs in the γ' phase. Figures reproduced from: (a) in Ref. [100] and (b) in Ref. [102].

increasing temperature (except for Ni-Cr at higher Cr content), and the largest decrease is observed for pure Ni.

2.2. Dislocation motion in γ' phase

To illustrate the creep strengthening effects of γ' phase, it is effective to consider the γ/γ' microstructure in Ni-SXs. As discussed above, the formation of γ' phase promotes the Ni-X (Al, Ti or Ta) bonds with forming $L1_2$ structures. The Burger's vector of a full dislocation is $a\langle 110 \rangle$ for $L1_2$ crystals, while the magnitude of that in the fcc structure of matrix phase is $a/2 \langle 110 \rangle$. In the fcc structure, the dislocation slips along the $a/2 \langle 110 \rangle$ (111) slip system, because $\langle 110 \rangle$ and (111) are the most closely spaced direction and lattice plane, respectively. As result, a single dislocation in γ phase cannot enter the γ' phase, since a full dislocation in the γ' phase is equivalent to two full dislocations in the γ phase and has higher lattice distortion energy.

The slip deformation in γ' phase is much different to that of γ . By combing two $L1_2$ crystals together, an interface named as anti-phase boundary (APB) formed. In the vicinity of the APB, a number of Ni-X bonds convert to Ni-Ni bonds and X-X bonds. The formation of APBs provides the channel for dislocations in the γ matrix to pass through the γ' phase. Generally, the dislocations in γ matrix travel in pairs across the γ' phase, with each functions as a partial dislocation in the γ' phase which are linked by an APB in the form of faulted strip, as shown in Fig. 5(a) [100]. The mechanism of dislocation pairs travelling across the γ' phase through APB is illustrated in Fig. 5(b). Two dislocations with a Burger's vector of $\mathbf{b} = a/2 \langle -101 \rangle$ travel in a sequence into the γ' phase. The first dislocation creates an APB on its path with locally converting the ordered bonds to disordered bonds, while the following dislocation sweeps the disordered area to eliminate the APB and restores the bonds to ordered $L1_2$ structures. To convert the ordered Ni-X bonds to disordered Ni-Ni and X-X bonds, the system needs a considerable increase of energy to form an APB, which is seen as the APB energy. Compared to common SFE of the matrix, which is about tens of mJ/m^2 , the APB energy is much larger. For Ni-Al structure, the energy to create the APB path is approximately to $150 \text{ mJ}/\text{m}^2$ [101]. However, in another mechanism, the movement of dislocation pairs travelling across the γ' phase is much different to the mode discussed above [102,103]. Also seen in Fig. 5(b), a first dislocation with a Burger's vector of $\mathbf{b} = a/3 \langle -1-12 \rangle$ travels across the (111) plane, that the stacking sequence of (111) planes is partly changed from ABCAB-CABC to ABCACABC to form a super lattice intrinsic stacking fault (SISF). Another dislocation with a Burger's vector of $\mathbf{b} = a/6 \langle 1-21 \rangle$ also travels across the same (111) plane to create a complex stack-

ing fault (CSF). Interestingly, if the two dislocations travel in pairs to across the (111) plane, the second dislocation converts the first one into an APB rather than forming the CSF. This is because the energy to form a CSF is relatively higher than that to form APB, where the formation of APB is promoted. Anyway, it is no matter the forming modes of APBs in the movement of dislocation pairs. The key point is the movement of dislocation pairs in the γ' phase requires the extra increase of system energy in the meaning of APB energy. Thus, the requirement of APB energy makes dislocations more difficult to move into the γ' phase referring to the great strengthening effects of γ' phase. As result, the strength of alloy is assumed to be proportional to the APB energy which significantly depends on the composition [104,105].

2.3. Creep mechanisms in the γ/γ' microstructure according to dislocation motion

The creep of Ni-SXs generally experiences three distinct stages according to the evolution of γ/γ' microstructure. Different features of creep exhibited in these three stages strongly correlate to the dislocation motion and the interaction between dislocations and γ/γ' microstructure. At the very stage of creep, under the assistance of external temperature and stress, dislocations gradually multiply in γ channels which are previously free of dislocations after proper heat-treatments. A short period of dislocation incubation is thus produced that dislocations percolate in those most stressed γ channels and the nearby γ/γ' interfaces to generate plastic flows in γ channels [10,106]. The onset of plastic flow was seen as the initiation of primary creep since the changed mechanical properties of γ matrix from previously elastic to plastic [107–109]. In primary creep, the strain rate of alloys firstly increases with the accumulate creep strains deriving from enhanced dislocation activities in γ channels, then decreases to a very low value. The decreased creep rate is because of the formation of dense dislocation networks at the γ/γ' interfaces, which are facilitated by increased dislocation activities in γ channels [110–112]. Dense dislocation networks help to restrict the movements of dislocations thus decreasing the creep rate in primary creep. In this situation, hardening effects provided by dislocation networks and softening effects resulted from the plastic dislocation flows work together to affect the creep of alloy. As the balance between hardening and softening achieved, the strain rate decreases to a very low level and keeps nearly constant giving rise to the onset of secondary creep [113–115]. Since the creep rate is very low, secondary creep accounts for most of the creep life of alloy, as shown in Fig. 6(a). However accumulations of creep strains during secondary creep would finally break

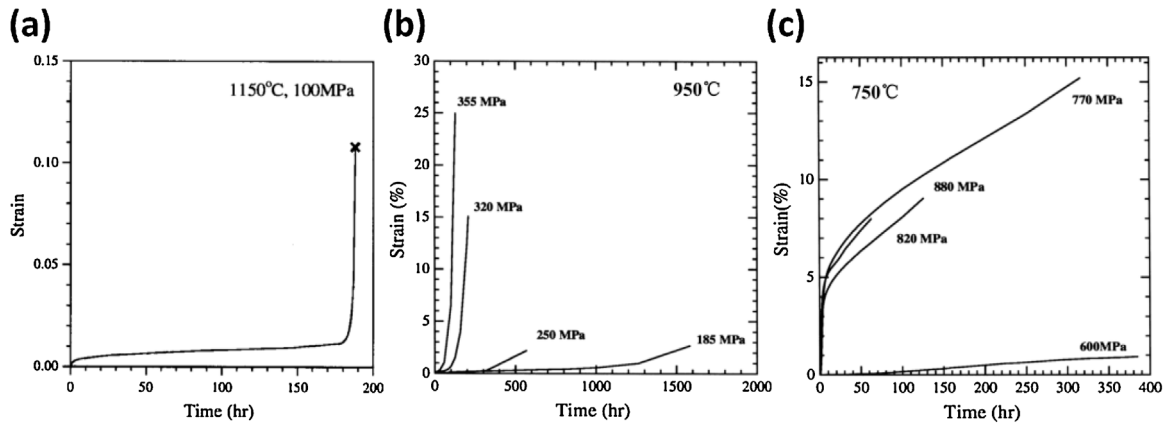


Fig. 6. Creep curves of Ni-SXs under different temperatures and stresses. (a) Curve of high temperature and low stress creep; (b) Curves of mid temperature and mid stresses creep; (c) Curves of low temperature and high stresses creep. Figures reproduced from: (a) in Ref. [131] and (b, c) in Ref. [133].

the balance of hardening and softening and lead to the consequent increase of creep rate in tertiary creep [116,117]. The accumulated creep strains significantly change the stress state and high level of stress concentration promotes the dislocation shearing events of γ'/γ precipitates to degenerate the γ'/γ microstructure. Therefore the strain rate in tertiary creep quickly increases with creep strain to facilitate the final fracture of alloy in a short time period (Fig. 6(a)).

The microstructural features displayed in different creep stages are always determined by the dislocation movements and their interactions with the γ'/γ microstructure. Therefore the evolution of γ'/γ microstructure is very sensitive to the temperature and stress applied on alloy which can greatly affect the dislocation motion [118]. Since dislocation activities involve the diffusion of solutes which is mainly influenced by temperature, a high temperature guarantees the activation of slip systems of dislocations [119,120]. At high temperatures (over 1100 °C), dislocations rapidly multiply to produce plastic strains in γ channels giving rise to the fast degradation of γ'/γ microstructure [121,122]. If the applied stress is not so high that most of dislocation movements are restricted in γ channels, the creep deformation is mainly caused by the accumulation of creep strains in the matrix giving rise to a long time period of secondary creep (Fig. 6(a)) [115,123]. The final failure also results from the accumulated creep strains in γ matrix which lead to a high stress concentration to promote the shearing of γ' precipitates [124–126]. However a high external stress can accelerate the creep process to decrease the creep life of each creep stage, especially the secondary creep. During a high temperature and high stress creep, the dislocation softening process always overweighs the hardening process with the assistance of the high stress. Thus only accelerating creep occurs in Ni-SX to exhibit a continuous tertiary creep feature [127,128]. Since no steady state creep is achieved, the creep life of high stress creep is much shorter than that of low stress creep (Fig. 6(b)). Under a relatively low temperature (around 750 °C), the creep mechanism is much different to those of high temperature creep. It was suggested that the dislocation density is low to produce abundant plastic flows within the γ'/γ microstructure at low temperatures [129,130]. Therefore minor creep strains accumulate to produce the obvious creep deformation under a low stress. And remarkable creep deformation only occurs to alloy under high external stresses [131,132]. In Fig. 6(c), during a 750 °C creep, stress lower than 600 MPa produced very small creep rates of alloy to result in a very long creep life, thus suggesting a stress threshold in low temperature creep [133]. In this situation, the creep deformation occurs by a great deal of shearing behavior in γ' precipitates because of the high stress makes dislocation pairs easily to cut into γ' precipitates [134,135]. Also a very small plasticity displayed in alloy during low

temperature and high stress creep, thus the creep curve is much steeper than those of high temperature and low stress creep as shown in Fig. 6(a).

2.4. Microstructural evolution during creep of Ni-SXs

At high temperatures, the originally cuboidal γ' precipitates occur to gradually degenerate to lose their regularity and form prolonged lath-shaped structures. This process is called rafting which is commonly exhibited in Ni-SXs with high volume fractions of γ' phase [109]. It was suggested that rafting strongly correlates to dislocation motion occurring in γ'/γ microstructure. And the distinct creep behaviors exhibited in different stages of creep should be related to the formation of rafted structures. Some researches denoted that the evolution of rafted structures is accompanied by the increase of dislocation movements in the γ'/γ microstructure as well as the evolution of dislocation networks at γ'/γ interface. For example, the formation of continuous rafts generally occurs in primary creep stage when dislocations gradually percolate in γ matrix. Meanwhile the increased dislocation activities also promote the formation of regular dislocation networks during primary creep [136–138]. Besides, the dislocation density keeps a nearly constant value during secondary creep and the morphology of rafted structures and dislocation networks also keep nearly invariant in this stage [66,89]. The destruction of interfacial dislocation networks is also along with the destruction of γ' rafts because of the increased dislocation shearing events during tertiary creep [107,139]. However these researches only illustrate the superficial relationships between the dislocation activities and the rafting process, the internal reason should be discussed further.

It is significant to point out that many minor ledges were formed at the γ'/γ interface (Fig. 7) during creep that was previously seen as the result of thermal-swelling of γ' precipitates during cooling of alloy [140]. But some recent researches based on nearly atomic scale characterization denoted those interfacial ledges could be formed by the intruding of dislocations which can give rise to the changed elemental distribution across the interfaces and cause the local dissolution of γ' precipitates [34,72,73]. Dislocation cores have been found at the tips of interfacial ledges with demonstrating the enrichment of γ phase elements, as shown in Fig. 8 [73]. The enrichment of γ phase elements, such as Re, Cr and Co at the dislocation cores can greatly increase the elemental inhomogeneity that gives rise to the expelling of γ' phase elements out of the γ' precipitate. The expelling of γ' phase elements would gradually change the local composition in front of the ledge tip, which is firstly the γ' phase, to make it gradually transformed to γ phase.

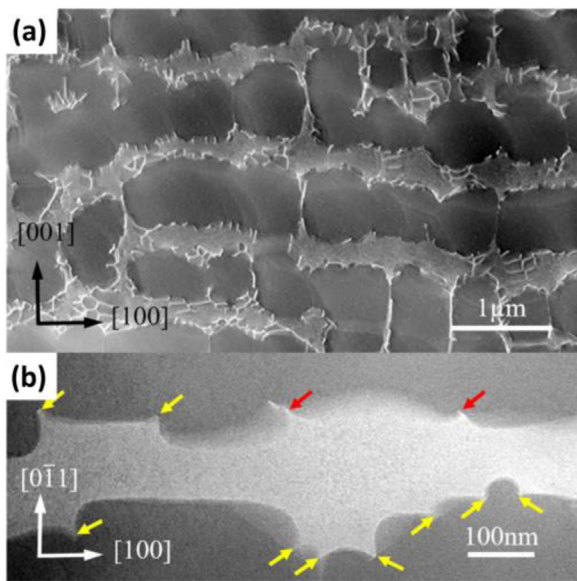


Fig. 7. Formation of interfacial grooves during creep of a Ni-SX under 140 MPa at 1120 °C. Adapted from Ref. [73]. (a) HAADF-STEM image of the superalloy after creep for 2 h observed along [010] axis showing many ledges formed at the interfaces; (b) Dislocations at the tip of each groove are indicated by an arrows.

Therefore, the γ' precipitate is locally dissolved with the intruding of dislocations at the γ/γ' interface [73]. The intruding dislocations at tips of interfacial ledges can help to connect the rafting process to the dislocation activities in γ/γ' microstructure. And some rela-

tionships were also displayed in the evolution of rafted structures according to the formation of interfacial ledges. On the one hand, interfacial ledges generally form with the formation of rafted structure in primary creep, when dislocations gradually percolate in γ channels with enhanced density, and the depth of these grooves increases with the coarsening degree of γ' precipitates [34]. On the other hand, once the continuous rafted structures form, the depth and density of interfacial ledges seem to be invariable during secondary creep [34,140]. Therefore we can expect that the rafting process is facilitated by these intruding dislocations. The formation of interfacial grooves which caused by intruding of dislocations must be related to the dissolution of γ' precipitates. But the real process of elemental transports with dislocations is not very clear yet. The hypothesis of 'pipe diffusion' could help to explain the enrichment of γ phase elements in dislocation cores at tips of interfacial ledges [141]. It was suggested that high density of dislocations helps to form diffusive channels at dislocation cores, which assist the elemental transport in γ matrix [74]. In addition, the formation of rafted structures needs the massive transports of γ' phase elements from γ/γ' interfaces normal to stress axis to parallel channels. But the real process of solute transports is still unclear.

The formation of rafted structures was shown to strongly correlate to the temperature and stress applied on alloy. On the one hand, a high temperature is the premise of rafting which guarantees the sufficient diffusion of alloying solutes. Also the rate of rafting should be positively related to the temperatures which facilitate the elemental diffusion as well as the corresponding dislocation movements. Apart from temperature, the orientation of rafted structures is very sensitive to the types of applied stress (tensile or com-

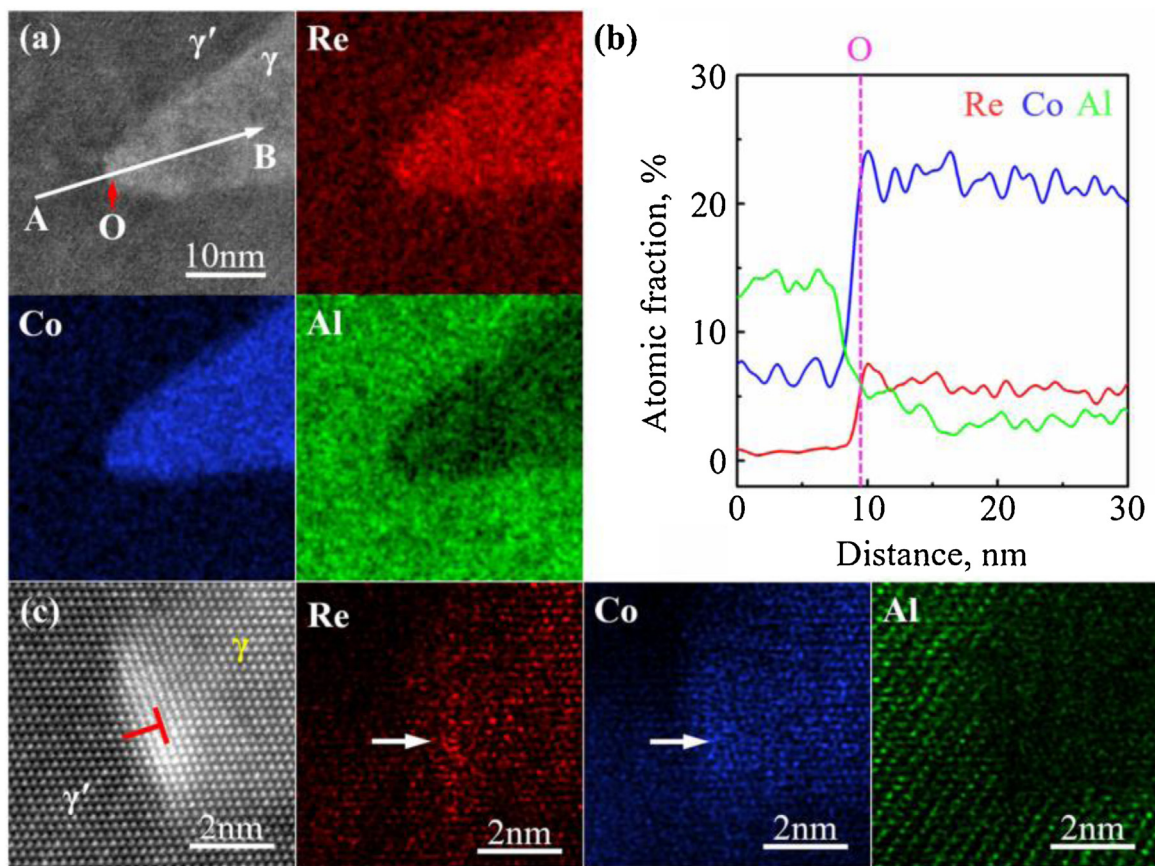


Fig. 8. EDX analysis of dislocation core at the tip of interfacial groove. Adapted from Ref. [73]. (a) EDX mapping of the groove tip showing the elemental inhomogeneity in region near to the groove tip. The arrow denotes the direction of line scan mode as shown in (b); (b) EDS data of line scan across the groove tip; (c) Nearly atomic EDX analysis of the dislocation core at the groove tip showing the enrichment of Re and Co.

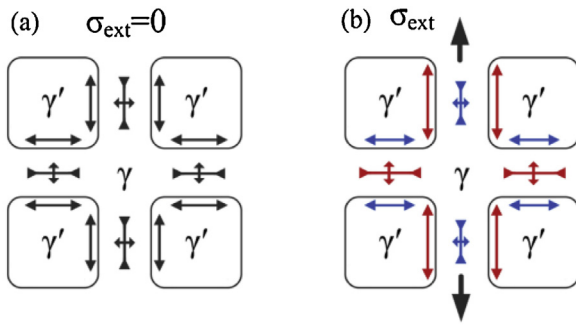


Fig. 9. Schematic figure of the stress distribution in the γ/γ' microstructure. Adapted from ref. [149]. (a) Internal misfit stress generates the stretched stress and compressive stress in the γ' and γ phases, respectively; (b) Tensile stress gives rise to the release of internal misfit stress in perpendicular γ channels but increases the stress state in paralleled channels.

pressive). Generally, a tensile stress applied in the $\langle 001 \rangle$ crystal orientation of Ni-SXs can give rise to formation of rafted structures perpendicular to stress axis (N-type), while P-type rafts paralleled to stress axis are produced by a compressive stress [142–146]. The directivity of rafts displayed in crept alloy has been concluded to result from the coaction of external stress and the internal misfit stress which derives from the mismatched γ/γ' interface [147,148]. As discussed before, most Ni-SXs have the negative lattice misfits at high temperatures, meaning that the lattice constant of γ phase is larger than that of γ' phase. In this situation, the γ' phase is internally stretched in absence of external stress, while γ phase is compressed because of the misfit stress, as shown in Fig. 9(a) [109,149]. A tensile stress gives rise to the release of misfit stress in paralleled γ channels, but increases the stress concentration in channels normal to stress axis (Fig. 9(b)) [66,149]. Therefore the coarsening of γ' precipitates mainly occurs in those paralleled channels because of the release of misfit stress, resulting in the formation of N-type rafts. Furthermore, it can expect that the reversed rafting behaviors occur in some positively mismatched alloys, indicating that the compressive stress and tensile stress produce the N-type and P-type rafts, respectively [144,145].

The evolution of rafted structures is closely related to dislocation movements occurred in the γ/γ' microstructure. To enhance the creep properties of alloy, it is necessary to restrict the movements of dislocations in the γ and γ' phases on the root cause. Alloying additions which help to increase the SFE of γ phase and/or the APB energy of γ' phase are still regarded as the important aspects in alloy design. For example, additions of Re, Co and Ru were shown to effectively decrease the SFE of γ phase because of their HCP lattice structure. And Ti and Ta in γ' phase can replace the Al atoms to form Ta-Ni and Ti-Ni bonds that greatly enhance the APB energy. These considerations are reviewed in detail in the next chapter. On the other hand, rafting was shown to be disadvantageous to the high stress creep properties because of the formation of rafted structures can greatly decrease the total area of γ/γ' interface. Therefore the difficulty for dislocations shearing the γ' precipitates is greatly reduced by rafting. Although some researches have suggested that the formation of rafted structures is beneficial to high temperature and low stress creep of alloy, which is mainly controlled by the gliding and climbing of dislocations in γ channels, because the formation of rafted structures can increase the route of dislocation gliding to enhance the creep resistance of the γ/γ' microstructure, an important expectation in designing Ni-SXs is to slow down the rafting process. Alloying additions such as Re which can stabilize the γ/γ' interface and reduce the coarsening kinetics could be a key research direction.

3. Composition evolution in Ni-based single crystal superalloys

An everlasting discussion is the fundamental effects of compositions on the mechanical properties of superalloys. Composition modulation is at the very early stage of alloy design and is easily operated by changing the types and quantities of alloying additions. However, it is also the most complicated step in alloy design since very slight change of any element can greatly influence the overall properties of alloys. Composition greatly determines the development of microstructures in superalloys. With the change of alloying contents, microstructural characteristics such as the size and volume fraction of γ' phase, the form of γ/γ' interface and the distribution of precipitated phase are consequently changed. From the micro perspective, the changed microstructures result in the variation of many structural parameters such as the lattice misfit, the SFE of γ matrix and the APB energy of γ' phase. These variations can change the mechanism of creep deformation and then influence the high temperature strength of alloys. Thus, modulation of alloying elements plays an important role in designing Ni-SXs. The common alloying additions in Ni-SXs are basically divided to two different classes due to the different strengthening mechanisms.

3.1. Addition of the γ' phase elements

The first class includes elements which are much larger than Ni, such as Al, Ti and Ta. These elements promote the formation of ordered γ' phase that help to enhance the creep resistance of alloys. By increasing the volume fraction of γ' phase to an appropriate level, the creep resistance would be enhanced. Compared to disordered γ phase, the γ' phase exhibits the ordered structure which is harder with greater resistance to dislocations. Most importantly, a great volume fraction γ' phase helps to impede the movements of dislocations giving rise to the substantial precipitating strength. However, increased volume fraction of γ' phase inevitably reduces the γ contents resulting in the reduction of total area of γ/γ' interface and width of γ channels. Since the interfaces play a key role in impeding the movements of dislocations and the γ channels provide the main plasticity of Ni-SXs, the creep strength will face to significant degradation while the total area of γ/γ' interface and width of γ channels reduce to the certain level [150,151]. As a result, the volume fraction of γ' precipitates needs to be control in an appropriate range, generally 60%–75% for most Ni-SXs [14]. And excessive additions of γ' phase elements would lead to the fast degradation of high temperature strength of alloy [150].

In a ternary Ni-Al-X system, the substitution behavior is depended on the atomic size of replaced atoms, that Al is replaced by larger solutes such as Ti and Ta. In the formation of ordered γ' phase, the additions of Ti and Ta promote the formation of Ti-Ni and Ta-Ni bonds. Compared to the Al-Al bond, the Ti-Ti and Ta-Ta bonds are much stronger which can enhance the APB energy. As shown in Fig. 10, the APB energy increased monotonically with the increase of Ti/Al ratio. The increase of APB energy results in the enhanced resistance of dislocation motion in γ' phase referring to the enhanced creep resistance [56,152]. Also nano-indentation analysis has shown that the hardness of the γ' phase increases with the addition of Ta and Ti, while the hardness of the γ channels remains nearly constant. Since Ti and Ta strongly partition to the γ' phase, the change in hardness of the γ' phase can be ascribed to the atom substitutions by Ti and/or Ta [153]. However, excessive addition of Ti would damage the high temperature creep resistance. In a Ti-Ni-Al system, element Al is replaced by Ti with an increase of lattice parameter of γ' phase. Thus the lattice misfit is enhanced and the γ' phase coarsening is inevitable with the degradation of creep resistance [56]. Besides, Ti eutectic structures which are hardly dissolved regularly form in casting process and destabilize

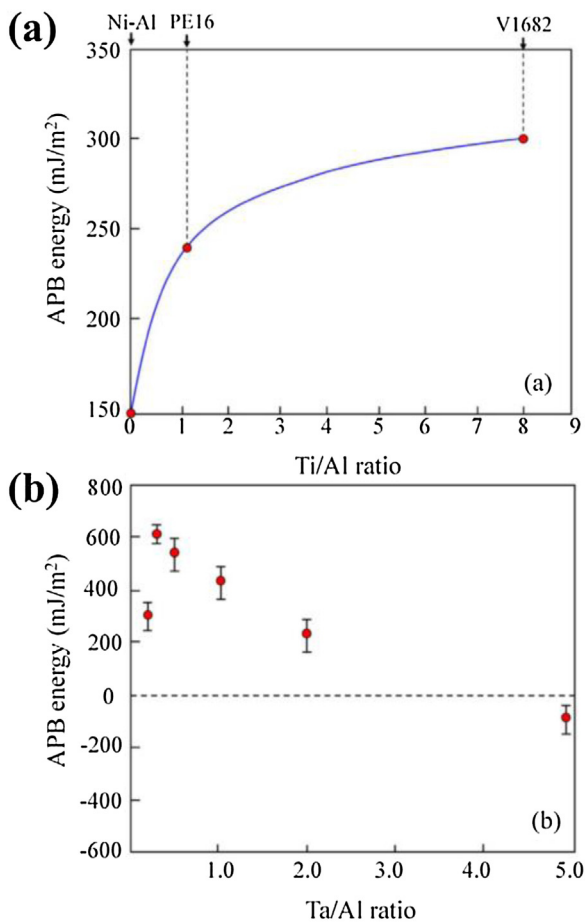


Fig. 10. Effects of Ti and Ta on APB energy of Ni-SXs. Adapted from ref. [84]. The effects of Ti/Al ratios (a) and Ta/Al ratios (b) on APB energy in Ni-Al-X systems.

the microstructures [154]. In more extreme condition, a high ratio of Ti/Ni promotes the precipitation of η -Ni₃Ti phase which provides the site for crack nucleation and propagation [53]. As results, Ti has been gradually abandoned from the third generation of Ni-SXs (Table 1).

By contrast to Ti, Ta is still extensively applied in single crystal alloys to enhance the creep strength and the resistance to oxidation and corrosion [155]. The effects of Ta on APB energy in a Ta-Ni-Al ternary system are much different. Fig. 10(b) shows that the APB energy increases greatly with a small ratio of Ta-Al and then decreases drastically with the continuous increase of Ta-Al ratio to a negative value after the ratio reached 5. A peak value of APB energy exhibits at a low ratio of around 0.2, which represents the supreme property [57]. Compared the influence of Ti and Ta on APB energy, Ta performs better in strengthening alloys [153,156]. Also, Ta was found to play a crucial role in stabilizing γ' precipitates. With addition of Ta, the enthalpy of forming γ' phase becomes more negative which suggests an increase of the stability of the γ' precipitates [157]. And a high content of Ta addition can inhibit the γ' phase from coagulating and growing during long exposure to high temperatures, thus restraining the coarsening of γ' phase [158]. The modulation of Ta/Al ratio serves as an important way in alloy design, which needs to be considered further. Recent studies showed that enhanced contents of Ta and Al can promote the formation of eutectic phase. At elevated temperatures, the large amounts of eutectic phase can deteriorate the structure stability and increase the TCP nucleation which greatly damages the creep strength of alloys [159,160].

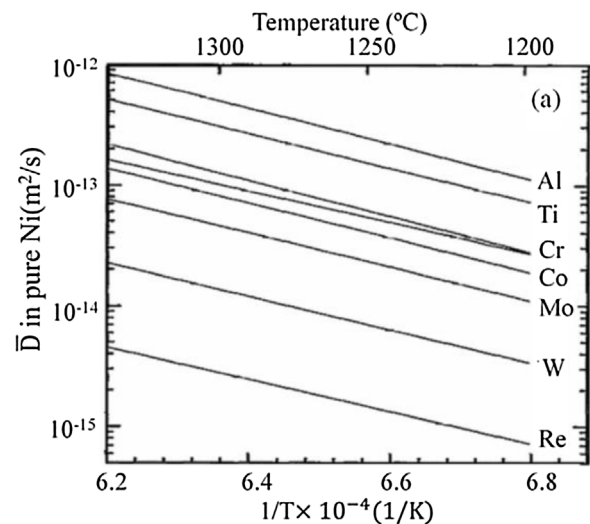


Fig. 11. Diffusion coefficients of some strengthening elements in Ni. Adapted from Ref. [161].

3.2. Addition of the γ phase elements

The second class includes Co, Cr, Mo, W, Re and Ru which form the γ matrix as solid solutions. These elements derive their creep-strengthening effects from their low mobility in Ni solvent to impede the diffusion of dislocations. The diffusion coefficients $\bar{D}$ of these alloying elements in Ni are given in Fig. 11. These elements exhibit gradually enhanced diffusion rate corresponding to the increase of temperature and are in the order of Al, Ti, Cr, Co, Mo, W and Re according to their diffusion coefficients. In conventional mechanism which involves the effect of misfit strain and the solute-vacancy binding interactions, larger atoms move slower according to the large binding energy. The atomic size of Al and Ti is much larger than that of Re and these large atoms should diffuse slower. However, it has been reported that in binary Ni-X systems Re diffuses several orders of magnitude slower than other 5d transition metals [161]. This is because solute atoms with atomic radii closest to that of Ni can display the slowest diffusion rate [162,163]. The origin of the diffusion trend lies in the characteristics of the d-states occupancy in the electronic structure of solute atoms [164]. The atomic radii of Re is only a little larger than that of Ni atoms, implying the very low diffusion rate of Re in Ni. A first-principle calculation has attempted to explain the slow diffusion of Re [165]. Calculated results showed that the positive solute-vacancy binding energy for Re is high which indicates that solute-vacancy binding is not favorable. From the electronic point of view, the origination of repulsive interaction between solute and vacancy possibly accounts for the low diffusivity of Re.

The slowest diffusion of Re in Ni implies an optimal creep performance of alloy with additions of Re. Besides, the coarsening of γ' phase requires the continuous diffusion of Re away from the γ/γ' interfaces, where a large amount of Re is segregated according to its low diffusion rate [40]. Thus, the growth of γ' phase is greatly retarded by the segregation of Re and then the creep resistance is enhanced [33,166]. Generally, the criterion for evaluating the high temperature performance of superalloys is the highest endured temperature of keeping 1000 h stability under a 137 MPa pressure. Fig. 12 demonstrates the temperature endurance of some typical Ni-SXs [5,84]. Profiting from the evolution of alloying contents, the endured temperature is significantly enhanced. For example, comparing to single crystal alloys CMSX-2 (0 Re) and CMSX-4 (3 wt% Re), the addition of Re in CMSX-10 is increased to 6 wt%, giving rise to the 20 and 4.6 time enhancement of creeping time, and the increase

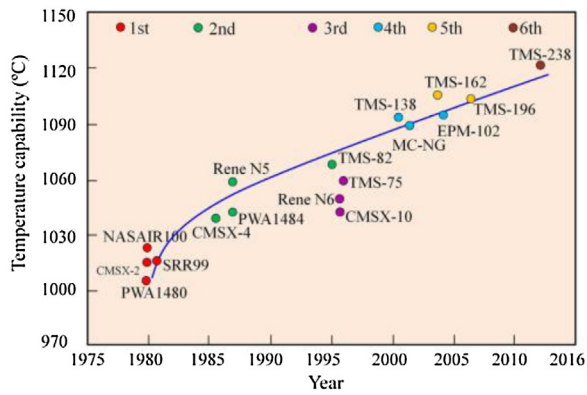


Fig. 12. High temperature capabilities of some typical Ni-SXs. Adapted from Ref. [84].

of temperature endurance about 30 °C and 60 °C [83]. Apart from Re, the gradually improved tendency of high temperature strength of Ni-SXs as demonstrated in Fig. 12 should also be attributed to the composition modulation of other major elements, especially the introducing of Ru in high generation Ni-SXs. By vertically comparing the compositions of TMS series Ni-SXs (Table 1), the contents of Ru increased from 0 (TMS-75, third generation) to 2 wt% (TMS-75, fourth generation), and then to 5 or 6 wt% in the fifth and sixth generations, giving rise to the great enhancement of temperature endurance. Some reasons may possibly explain the effects of Ru, such as restraining the TCP phase and stabilizing the microstructures, while detailed discussion can be seen in Section 3.3. However the increased contents of Re and Ru in new Ni-SXs would inevitably decrease the total contents of other γ phase elements to keep the volume fractions of γ' phase in a proper range. The trend of composition evolution in Ni-SXs also shows a gradual decrease of proportion of element Co and Cr (Table 1). The total additions of element Co and Cr have reduced from over 15 wt% in early superalloys to around 10 wt% in new generations. Compared to Cr, Co additions are more favorable in alloy because its effect on reducing the SFE of γ phase which derives from its intrinsic HCP lattice structure. A low SFE helps to restrain the dislocation movements thus enhancing the creep properties of alloy [99]. But it was found that high level of Co addition can decrease the solvus temperature of the γ' phase and decrease the equilibrium volume fraction of γ' phase, leading to the increase in the partitioning ratios of TCP-forming elements Cr, Mo, Re, and W, especially for Re [167,168]. As a result, the elemental super-saturation of these TCP-forming elements is increased and the phase stability is reduced, promoting the formation of TCP phase [169,170]. Besides, high contents of Cr are disadvantageous to the creep properties of alloy even it can provide a large structural stability at high temperature. High Cr contents will lead to a decline of hot workability because of the formation of Cr-rich bcc phase, which has poor plasticity at high temperatures [171]. Furthermore, refractory elements Re, W and Mo have been extensively employed in superalloys, because of their contributions to enhance the solvus temperature and creep strength, as well as stabilizing the microstructures at elevated temperatures. To quantify the effects of these refractory elements on creep strength, Fleischmann et al. [172] had prepared several single-phase alloys based on Re, W and Mo with the composition of nickel solid-solution matrix in single crystal forms. The special single-phase structure ensures very clean data because no grain boundary or particle strengthening effects interfere with the solid-solution hardening. The results of creep test shows that Re is much more effective for matrix strengthening than either W or Mo. Thus, a strong creep-strengthening effect can be attributed to Re [173–175], which diffuses more slowly in Ni comparing to other alloying elements.

3.3. Effects of Ru additions on TCP phase forming and creep properties

Excessive additions of refractory elements will be unfavorable to the high temperature strength of superalloys. The great segregation of these elements can eventually promote the precipitation of brittle TCP phase which are detrimental to the creep resistance of alloys. As TCP phase formed by elements co-segregation gives an upper limit for the addition of refractory elements in modern Ni-SXs, deeper considerations of the influence of certain alloying elements on the partitioning behavior of other elements are critical. The addition of element Ru was started at the fourth generation and was shown to stabilize the microstructures with respect to suppress the precipitation of TCP phase [176,177]. Because Ru strongly partitions to the γ matrix, the enhanced stability of TCP phase arises from the reduced partitioning coefficients of refractory elements in γ matrix, which is defined as “reversed partitioning” [35,36]. As a result, the nucleation of TCP phase is suppressed and the growth rate of TCP phase is decreased [24,178]. In Fig. 13(a), superalloy Astra1-21 contains 1 at.% Ru at the expense of Ni while Astra1-20 is Ru-free. It is clear that the number density of TCP precipitates is much lower in the first alloy, indicating that TCP phase nucleation is impeded by Ru addition [179,180]. Also it is showed that a large amount of Ru can increase the solubility of other alloying elements, such as Al, Re and Cr in the γ phase [181]. As a result, the ejection of these elements out of the γ matrix requires a higher temperature referring to enhanced phase stability [182,183]. And with increasing solubility, the enhanced level of strengthening elements in γ matrix such as Cr can decrease the SFE of the γ phase resulting in larger lattice misfit and denser interfacial dislocation networks, in turn to increase the creep resistance of alloys [70,71]. In addition, Ru gives rise to the formation of TCP phase in a discontinuous type which transforms the supersaturated and unstable γ/γ' microstructure in a condition close to equilibrium, and thus the γ' phase is stabilized [183]. Furthermore, the formation of TCP precipitates undergoes sequential phase transformation. Ru addition can retard the phase transformation leading to decreased equilibrium volume fraction of TCP phase [184]. As a result, the precipitation of TCP phase is greatly restrained, referring to the enhanced high temperature strength [184,185].

However, the role Ru playing in such behavior of reversed partitioning is still under discussion and the underlying mechanism for this phenomenon is still unclear. Volek et al. [186] considered that reversed partitioning would be the synthetic effects between Ru and other alloying elements instead of the single addition of Ru. In their experiments, the obvious reversed partitioning can occur only if very strong changes of concentration of several elements rather than a small change of one particular element. And in some other discussion, the additions of Ru even promoted the formation of TCP phase and damaged the strength of alloy which are in contrast to previous studies [187,188]. The computational results showed the strong interaction between Re and Ru originating from d–d orbital hybridization was the reason referring to the pinning effect of Ru to Re atoms with substituting other atoms around Ru [187]. The substitution of Re to other atoms is expected to enhance the partition degree of other γ phase elements particularly to those other TCP-forming elements, such as Cr, and increase the nucleation of TCP phase [187]. In another recent study [188], the addition of 3 wt% Ru caused the reverse partition of alloying elements (Cr, Co, Al and Ta), but promoted the TCP precipitates in alloy with high Cr content. This discrepancy may result from the high-content Cr (~9 wt%), which greatly enlarged the super-saturation of Cr and Co in γ matrix and then promoted the precipitation of TCP phase. Besides, since TCP precipitates are surrounded by γ' envelopes, addition of Ru will hinder the partition of refractory elements to the γ phase with shifting the γ composition toward that characteristic of γ' phase

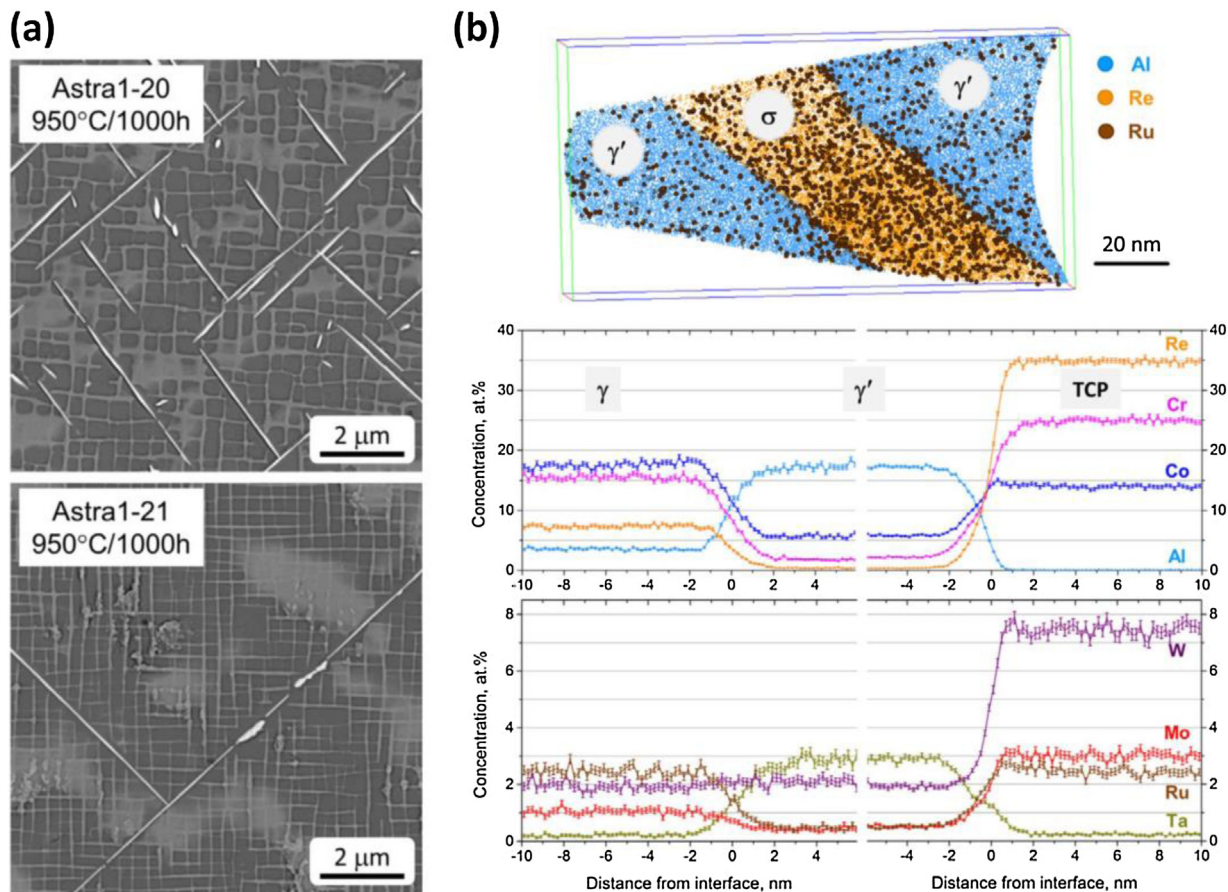


Fig. 13. Effects of Ru addition on TCP precipitation and elemental segregation of refractory elements. (a) SEM micrographs of alloys after isothermal annealing at 950 °C for 1000 h; (b) An APT elemental map of a plate-like TCP precipitate surrounded by a γ' envelope in the annealed Astra1-21 alloy. Accumulation of Cr, Mo, W and Re in TCP phase is visible. Figures reproduced from: (a) in Ref. [179] and (b) in Ref. [180].

and hence locally destabilize the γ phase, as shown in Fig. 13(b) [180,189]. Thus, during tests below and at the temperature corresponding to the peak strength, the yield strength of Ru containing alloys is lower than those alloys without Ru [176]. In addition, Latief et al. [190] found that Ru addition can indirectly promote the precipitation of TCP phase between the aluminum coat and substrate. Since the inter-diffusion of Ru, Al and Ni, the solubility of some of the refractory elements decreased in the diffusion zone. Thus the Ru indirectly increased the super-saturation of Re and Cr in the γ matrix, and consequently promoted the precipitation of TCP phase.

Anyway, Ru additions in new generation Ni-SXs undoubtedly affect the formation of TCP phase and the high temperature strength [191–193]. At mediate temperatures, Ru additions can reduce the SFE of γ phase and this is thought to have a significant contribution towards its strengthening benefit [191]. The high temperature creep properties of Ru-contained alloys are respected to the overall consideration including the effects of Ru on γ' -volume fraction, γ' -size and coarsening rate, as well as the lattice misfit and the γ/γ' partitioning coefficient. It was found that the reduction in γ' volume fraction upon the addition of Ru appears to be the principal cause of its diminishing strengthening contribution at elevated temperatures [191]. As stated above, Ru additions greatly influence the partitioning ratios of alloying elements, in turn to change the γ/γ' lattice misfit. In most cases, the γ/γ' lattice misfit became larger negative with Ru additions. The larger misfits can prolong the primary stage by increasing the strength of γ phase to impede the dislocation movements and reducing the initial γ' size, which slows down the γ' phase coarsening [192]. Tan et al. [194] systematically investigated the effects of Ru addition on the creep

strength of Ni-SXs at very high temperatures. Alloy with 4Ru addition had the superior creep properties and its creep life was three times longer than that of Ru-free alloy (Fig. 14(a)). Also the minimum creep rate was markedly decreased due to the additions of Ru (Fig. 14(b)). Since Ru additions can increase the lattice misfits, the strong elastic interaction of dislocations promoted the interfacial dislocation networks to become denser and more uniform during the secondary creep stage (Fig. 14(c–e)) [92]. The denser dislocation networks further impeded the dislocation motion which is the immediate reason for the increased creep life in the secondary creep stage and the superior creep properties.

3.4. Re effects

We now know that Re segregates strongly to γ phase where dislocation movements mainly take place, and the additions of Re can retard the movements of dislocations giving rise to an overall enhancement of creep resistance and rupture life, called “Re effects” [195,196]. A reasonable assumption is that Re could increase the activation energy of dislocation in the γ phase, and thus the thermal motivation of dislocations is restrained. It was suggested that alloying with Re results in directional and incompressible Ni–Re bonds, owing to the nature of electronic bonding and to the high density of unpaired electrons. Indeed, in a binary Ni–Re system, a very minor addition of Re can double the activation energy to previous value [119]. However, the specific mechanism of Re acted on dislocations is not clear until now. In an early theory, Re diffuses very slowly in the γ phase and segregates together to form clusters with the movements of dislocations. The formation of Re cluster further

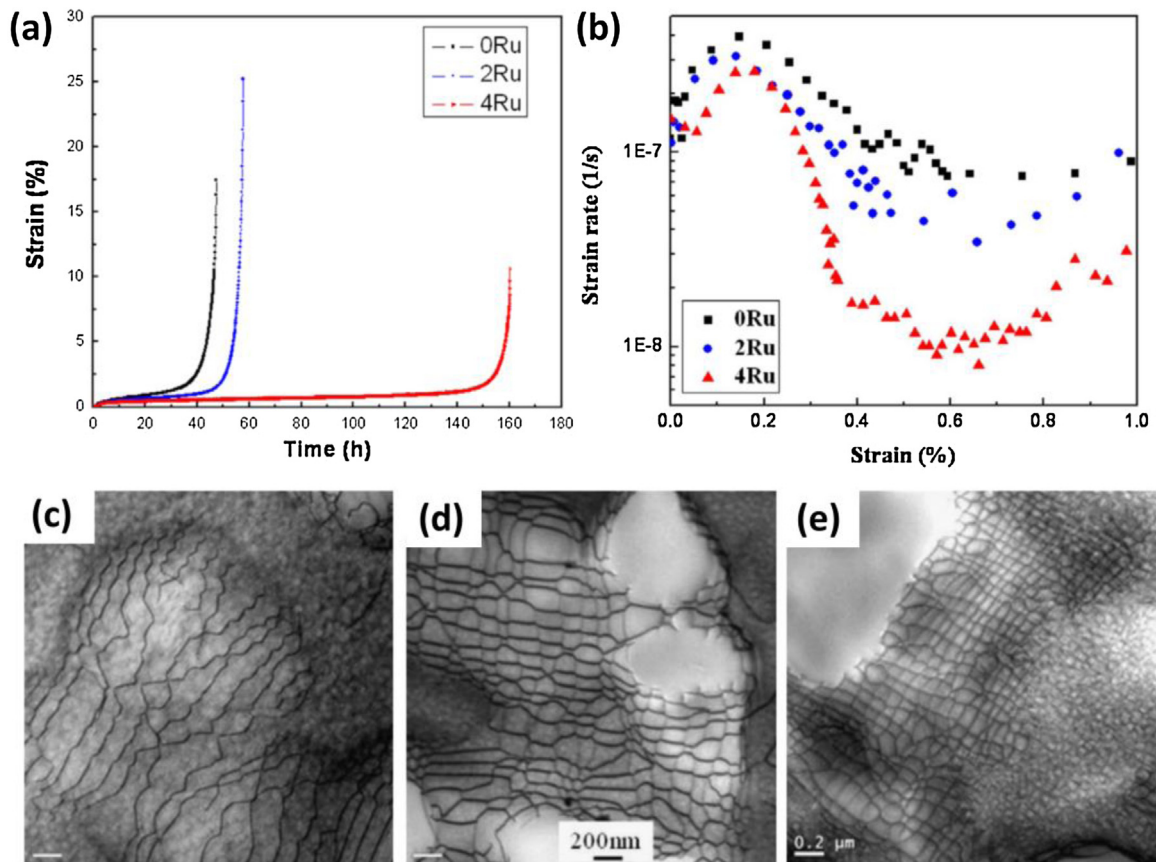


Fig. 14. Effects of Ru addition on creep properties of a Ni-SX. Adapted from ref. [194]. (a, b) Creep curves for the three alloys at 1150 °C /100 MPa; (c–e) Morphology of the interfacial dislocation networks in the alloy; (c) 0Ru, (d) 2Ru and (e) 4Ru after thermal exposure at 1150 °C for 500 h.

hinders the movements of dislocations in γ phase, inhibits the rafting process of γ' phase and increases the lattice misfit [197]. The Re clusters theory seemed very prospective to illustrate the retarding effects of Re on dislocations and many researches have conducted on this theory to prove its validity [198–200]. The APT technology is effective to analyze chemical compositions and distributions of materials in atomic level. Surely, the ladder behavior of Re was demonstrated in some APT researches which proved the regional enrichment of Re, as shown in Fig. 15. However, these results were weakly confirmed to the existence of Re cluster by simply displaying the fluctuation of Re concentration in the γ matrix. And it was found no pronounced dependence of Re concentration to the inter-diffusion coefficients in the Re-Ni binary system [201]. Also, even though Re clusters are certainly existed, their underlying effects on dislocation motion are still under discussed.

The argument that the presence of Re cluster in matrix is the main reason for the great increase of creep resistance in superalloys is still under discussion and recently many researchers hold the opposite view [172,202]. Mottura et al. [203] considered the ladder behavior of Re in the APT spectrum as the random variation in solute atoms by taking account of the positional scatter of the atoms in γ matrix. In an extended X-ray absorption research, the simulation $\chi(R)$ spectrum of a single Re absorbing atom embedded in a Ni-FCC lattice exhibited a double-peak behavior under the premise of existing Re clusters, but it was not shown in experimental results, as shown in Fig. 16(a) [202]. In another density functional theory simulation, strong negative binding energy existed in the Re-Re neighbor pairs which greatly restrains the assembly of Re atoms, as shown in Fig. 16(b) [204]. Thus, the formation of Re clusters is unlikely promoted in nature. Indeed, solid evidences which indicate the existing of Re cluster are still lacked [172]. And the influence

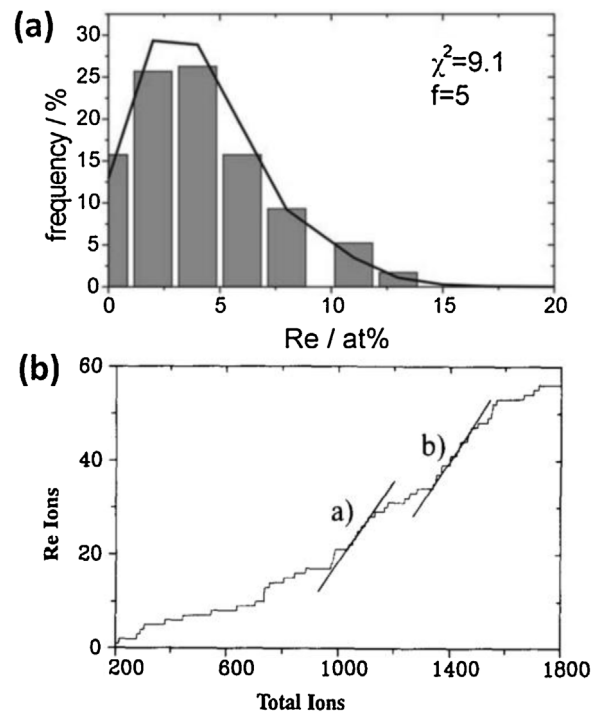


Fig. 15. Diagrams of Re distribution in γ phase showing the fluctuation of contribution and ladder behavior. (a) Frequency distribution of the Re concentration represents the binomial distribution expected for a statistically random distribution of Re atoms; (b) Re ions versus total number of detected ions. Figures reproduced from: (a) in Ref. [200] and (b) in Ref. [199].

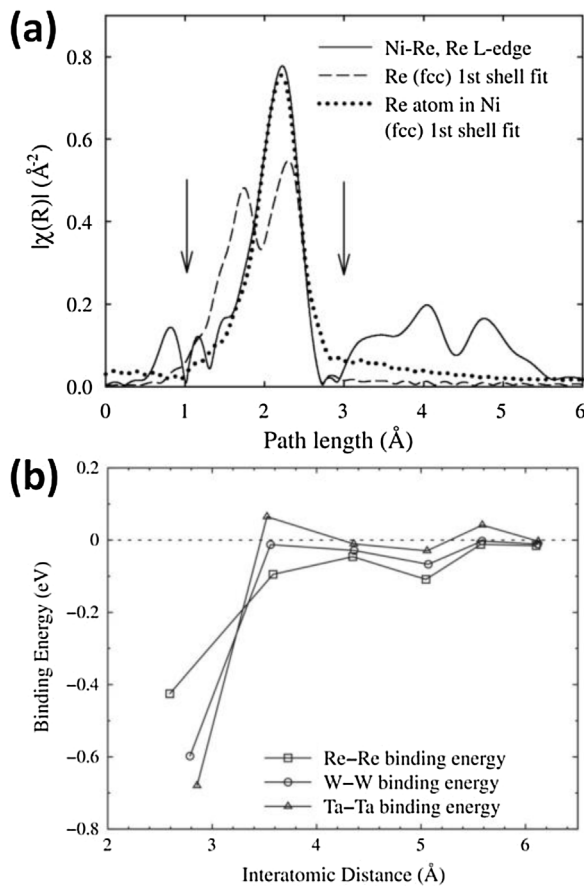


Fig. 16. Simulation works of Re-Re bonds against the hypothesis of Re clusters. (a) $\chi(R)$ spectra obtained from Re L-edge of a Ni-Re sample compared to fits obtained from a Re atom with 12 Re neighbors and a Re atom with 12 Ni neighbors. Arrows show the fitting range; (b) Binding energies between Re-Re, W-W and Ta-Ta pairs in FCC Ni supercell as a function of X-X separation. Figures reproduced from: (a) in Ref. [202] and (b) in Ref. [204].

of Re-Re interactions on the energetics of diffusion process is still unknown. More in-situ observations of dislocation motion according to the possibly existed Re cluster are required.

Anyway, the addition of Re surely improves the creep behavior of γ phase and greatly enhances of high temperature strength of Ni-SXs. Note that the coarsening of γ/γ' microstructure does depend on not only Re diffusion, but also the diffusion of γ' phase elements in the γ matrix which additionally contributes to the creep deformation. The interactions of Re with other elements also need to be considered. The co-segregation behavior of Re with other alloying elements was found at the γ/γ' interface after the creep deformation of alloy [29,34]. It provided new sight that the creep resistance of alloys might derive from the combined action of Re and other alloying elements instead of the simple addition of Re. In other words, other elements might assist to form the Re cluster and they together work on strengthening the γ/γ' interface. It was also found that Re could increase the segregation of other alloying elements. With the addition of Re, the W content increased in the γ phase, while the Cr content decreased, which is considered as the reason that Re promotes the precipitation of TCP phase [205]. In summary, the formation of Re cluster is still a controversial postulate. The explorations of so-called “Re effects” to confirm the real mechanism of Re in strengthening the Ni-SXs are still fascinated. “Re effects” characterizes in many aspects, including enhanced segregations of other alloying elements, distorted dislocation networks, increased misfits and revised rafted structures [175]. In the process of creep deformation, the enrichment of Re appears near to the

γ/γ' interface resulting from the slowest diffusion rate of Re in the γ matrix [206]. It was concluded that Re is continuously expelled into γ matrix during the growth of γ' phase, since the region close to γ/γ' interface exhibited the pileup of Re atoms in γ matrix near to both large primary γ' rafts and also small secondary γ' precipitates. The enrichment of Re solute was confirmed in another research in which the distribution of alloying elements near to the γ/γ' interface was measured by 3D-APT (Fig. 17(a)) [207]. In addition, the aggregation of Re solutes near to γ/γ' interface may affect the inter-diffusion of other alloying elements across the γ/γ' interface. Ge et al. [208] analyzed the elemental distribution of crept Ni-SX and the results showed the pile up of Ni and Ta solutes at γ/γ' interface ahead of the enrichment of Re, as shown in Fig. 17(b–d). This phenomenon implies that enrichment of Re may restrain the diffusion of other alloying elements, and then retard the coarsening of γ' precipitates to enhance the creep resistance of alloys.

In the initial stage of creep, the deformation is mainly driven by dislocation motion in the γ matrix referring to the glide and climb process, and the movement of dislocations is naturally an inter-diffusion process of alloying elements. The enrichment of Re solutes at γ/γ' interface can retard the inter-diffusion and enhance the segregation of other alloying elements, and then the dislocation motion is restrained. The detailed interaction between dislocations and Re solutes at the γ/γ' interface was manifested in an HAADF-STEM research [33]. As shown in Fig. 18(a), a V-shape protrusion at the γ/γ' interface forming by the edge dislocations is exhibited in the crept alloy. Besides, the elemental distribution map in the protrusion region showed a co-segregation of Re, Co and Cr at the tip of protrusion (Fig. 18(b)) which was considered as the reason of the formation of interfacial protrusion. Since the enrichment of Re retards the inter-diffusion of Co and Cr, the movements of dislocations are restrained to form the distorted dislocation networks. In another in-situ tensile observation, the fracture surfaces of Ni-Al and Ni-Al-Re-Ru alloys were nearly parallel to (010) planes and that of the Ni-Al-Re alloys was nearly parallel to (1 – 1 1) plane. It was expected that the addition of Re can reduce the SFE of Ni-Al-Re alloy to form stacking faults on the (1 – 1 1) plane which inhibits the cross-slip of other dislocations and forces dislocations to slip along the (1 – 1 1) plane [32]. In the later stage of creep, the deformation feature is the $\langle 110 \rangle$ super-dislocation shearing into the rafted γ' phase and slipping on (111) planes. However, evidence showed that the minor addition (2 wt%) of Re promotes the formation of Kear-Wilsdorf (KW) locks within the γ' phase in middle temperature range, as shown in Fig. 19 [209]. And the dislocations in KW locks under thermal activation may be re-activated for slipping on (111) planes to release the KW locks at the elevated creep temperature. Thus the addition of Re can stabilize the rafted structures with increasing the creep resistance at medium temperatures.

3.5. New discussions of Mo and W

Refractory elements Mo and W have been extensively employed in Ni-SXs for decades. As solid solution element, Mo strongly partitions to the matrix as a solid solution strengthener. Like other γ phase elements, Mo changes the γ/γ' lattice misfit and retards the coarsening of γ' precipitates to enhance the creep strength of alloys. But Mo was also confirmed to promote the formation of TCP phase which damages the high temperature strength of alloys [81,210,211]. As a result, Mo has been sparingly applied in a minor content of most commercial Ni-SXs. However, the successful developments of some advanced alloys have brought Mo to the attention of researchers again. One of the most representative superalloys is TMS-162, a fifth generation Ni-SX, contains 3.9 wt% Mo and exhibits the superior creep strength [17]. In view of the greatly enhanced high temperature properties, an extensively accepted statement is that Mo enlarges the negative lattice misfit of γ/γ' structures

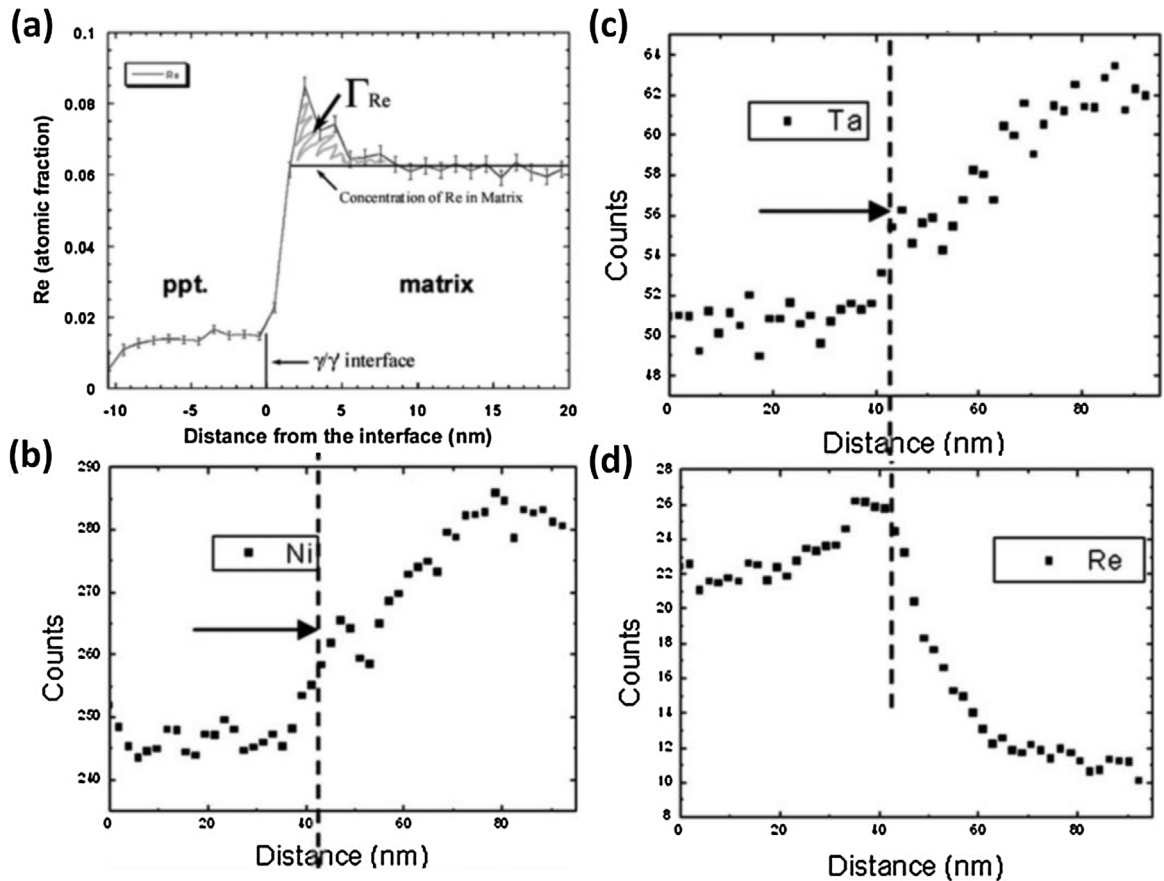


Fig. 17. Co-segregation of Re and other elements at the γ/γ' interface. (a) Re concentration as a function of distance with respect to the γ/γ' interface exhibiting enrichment of Re in the γ matrix near to the γ/γ' interface; (b–d) Pileup of alloying elements Ni (b) and Ta (c) near to the γ/γ' interface ahead of the enrichment of Re solutes. Figures reproduced from: (a) in Ref. [207] and (b–d) in Ref. [208].

and gives rise to the formation of dense γ/γ' interfacial dislocation networks during high temperature creep [67–69]. In Fig. 20(a and b), Mo contents in alloy I and alloy II are 1 wt% and 2 wt%, respectively. After high temperature creep, the dislocation networks are denser in alloy II with smaller average dislocation spacing and the resulted creep life of alloy II is longer [68]. Besides, Mo addition was found to positively influence the partitioning behavior of other alloying elements, especially for Re [69]. The enrichment of Re, W and Cr in the γ matrix caused by Mo addition leads to a further increase of the γ lattice constant with negatively increasing the lattice misfit. At low temperatures, the enlarged lattice misfits contribute to stabilize the microstructures and enhance the creep life. But the enhanced partition of these refractory elements also improves the opportunities for forming TCP precipitates at elevated temperatures (Fig. 20(c and d)) which greatly degrades the high temperature stability. Furthermore the increasing addition of Mo was shown to suppress the γ' phase coarsening. It was shown that the activation energy for coarsening can decrease with the increase of Mo content, and thus the coarsening rate decreased [212,213].

Different partition behavior is exhibited to element W. Although as solid solution strengthener, W partitions preferentially to the γ' phase [214]. And with increasing content, the partitioning coefficient of W to γ' phase decreased [215–217]. The decrease of partitioning coefficient was caused by increased W content in the γ matrix, which was not balanced by a similar increase in the γ' precipitates [218]. APT studies in Fig. 21(a and b) indicated that the partitioning of W atoms to the γ' phase is reversed in favor of the γ phase due to Ta additions. First-principle calculations of the sub-

stitutional formation energies indicated that W and Ta atoms share the Al sublattice sites of the γ' phase and the substitutional formation energy of Ta in the γ' phase is greater than that of W (Fig. 21(c)). Also, the calculated binding energies of Ta-W and W-W dimers in the γ and γ' phase indicate that the interatomic interactions of Ta-W dimers are repulsive in the γ' phase, while the W-W dimers are attractive in the γ phase (Fig. 21(d)). This implies that the interaction between Ta and W distract W from the γ' precipitates into the γ matrix [219].

On the contrary to the γ and γ' phase elements such as Al, Ta, Mo and Re, which specifically partition to a single phase, in most cases W atoms are distributed in both the γ' precipitates and γ matrix to influence the segregation of other alloying elements and in turn to change the microstructures in the both sides across the γ/γ' interface [207,220]. And the lattice parameter misfit is particularly sensitive to variations in the concentration of W on both sides due to its large atomic size [221,222]. In γ' phase, like other γ' phase elements the enhanced content of W can increase the γ' solvus temperature even the increased tendency is not as strong as Ti and Ta. This means that the enrichment of W in the γ' phase is connected to a stabilization of γ' to higher temperatures [223,224]. Furthermore, W in the γ matrix was shown to behave like Re which stabilizes the γ/γ' interfaces to inhibit the crack propagation at low temperature, while at high temperatures W can improve the ductility of alloys [225]. Also, large amount of W promotes the formation of TCP phase in the matrix which greatly damages the high temperature strength. And Ti additions was shown to restrain the partition tendency of W atoms to γ matrix and the formation of TCP phase [226,227].

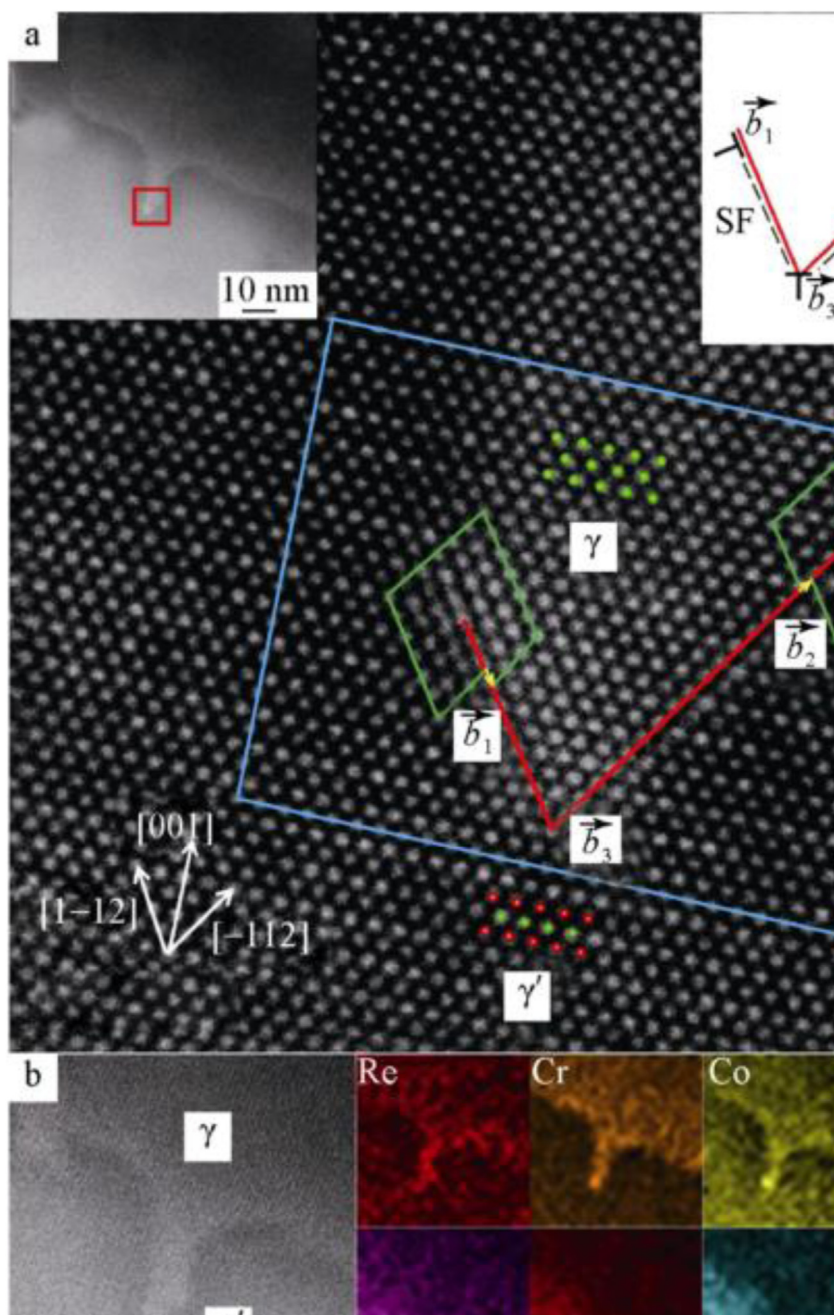


Fig. 18. V-shape protrusion at the γ/γ' interface forming by edge dislocations in a crept Ni-SX. Adapted from ref. [33]. (a) HAADF-STEM images of dislocation structures at tip of a V-shaped protrusion; (b) Elemental distribution maps in region near V-shaped protrusion.

4. Conclusion and perspective

The creep properties are greatly decided by the compositions and microstructures in Ni-SXs. Over ten kinds of alloying elements are added in modern commercial Ni-SXs and each of the elements takes different part in strengthening alloys. Among all the alloying elements, Re plays a special role in the enhancement of high-temperature creep properties. The evidence has indicated that the great increase of creep resistance in new Ni-SXs is undoubtedly affected by additions of Re. Many researches were conducted to find out the real effects of Re on strengthening the γ/γ' microstructure and to make clear the definition of “Re effects”. The previous assumption of “Re cluster” is suspected because of the lack of direct observation of the large area of Re atoms aggregation. Another recent explanation seemed to be plausible that Re can segregate

to the γ/γ' interfaces to retard the inter-diffusion of elements and the coarsening of γ' precipitates. However, it is still lack the solid evidence to confirm statement because of the limited dynamic characterization of γ/γ' interfaces in atomic-scale. Elements W and Mo once were restricted to be used in Ni-SXs, which are recently discussed again according to their special effects on the distributions of other elements and the increase of γ/γ' lattice misfits. It was found that additions of W and Mo are advantageous to the creep properties of alloys, but the reasons are still under discussed. The use of refractory elements such as Cr, Re, W and Mo can increase solvus temperatures of Ni-SXs. However, excessive additions of these elements will give rise to the precipitation of TCP phase, which is detrimental to the creep properties of alloys. The enrichment of refractory elements in TCP phase makes it very brittle under external stress. And the formation of TCP phase inevitably

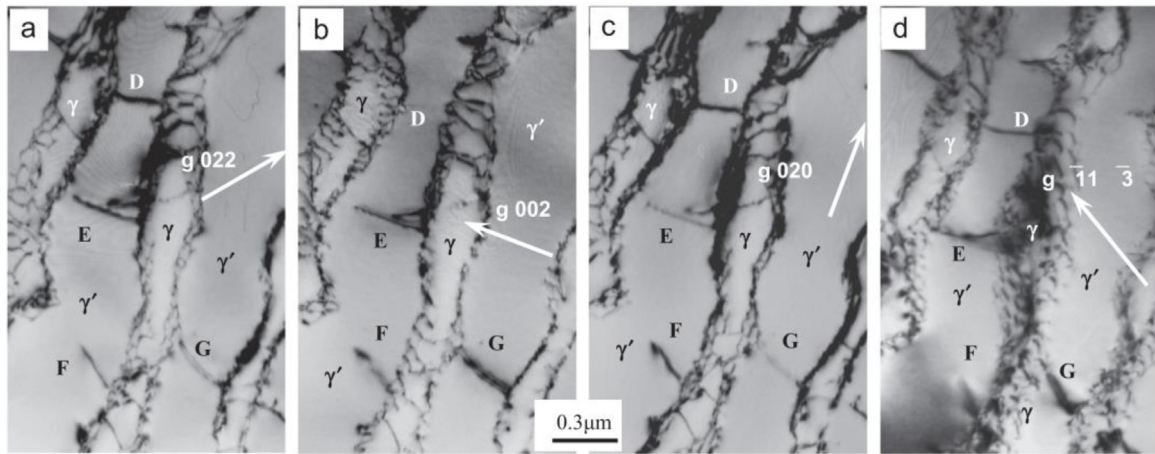


Fig. 19. Dislocation configurations within the rafted γ' phase in 2% Re alloy crept for 155 h up to fracture at 980 °C/300 MPa. Adapted from ref. [209]. (a) $g = 022$, $B = [100]$, (b) $g = 002$, $B = [100]$, (c) $g = 020$, $B = [100]$ and (d) $g = -11-3$, $B = [30-1]$. It is a reasonable consideration that the dislocations D and E on (100) planes are KW dislocation locks which originate from the cross-slipping of the dislocations from (111) plane to (100) plane.

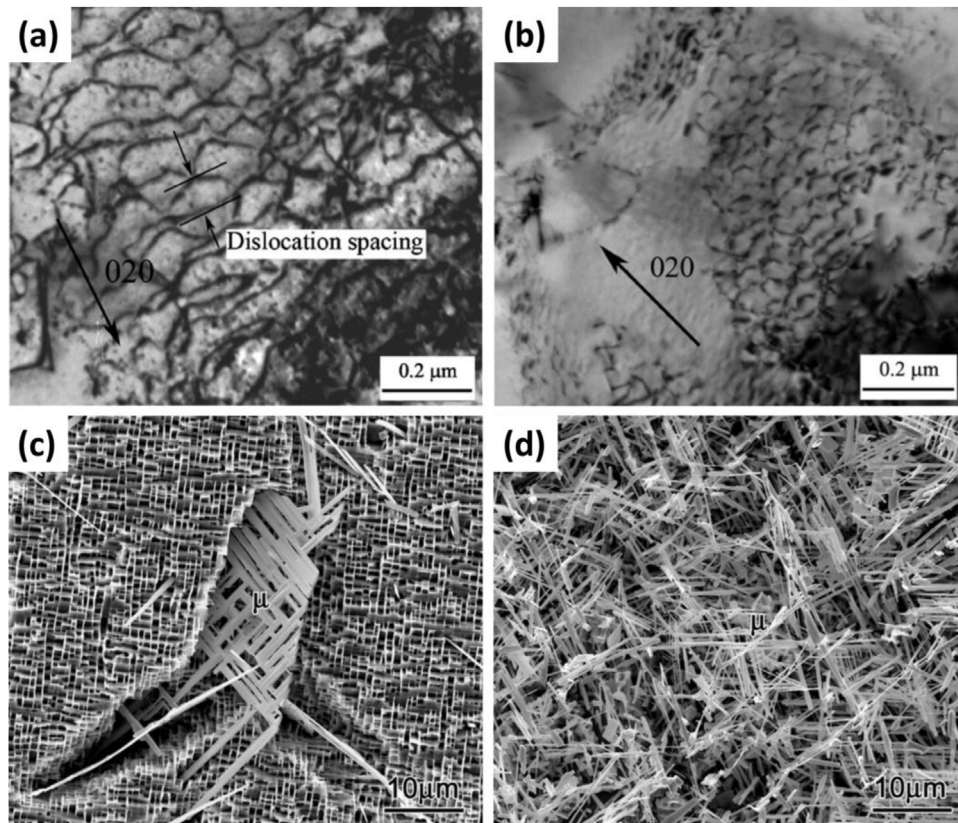


Fig. 20. Effects of Mo additions on dislocation networks and TCP precipitation. (a, b) TEM pictures showing the dislocation networks at γ/γ' interfaces of alloy I (a) and alloy II (b); (c, d) SEM images of the TCPs in alloys after exposing at 1100 °C for 200 h. The Mo contents in (c) and (d) are 1.5 wt% and 2.5 wt%, respectively. Figures reproduced from: (a, b) in Ref. [68] and (c, d) in Ref. [69].

consumes the strengthening elements and destroys the original regular cross-networks consisting of γ matrix and γ' precipitates. Thus the vicinity of TCP phase always serves as nucleation site of micro-cracks and propagation of micro-cracks is generally along the TCP phase. The big problem brought by TCP phase leads to the addition of Ru which begins at the fourth generation of Ni-SXs. Although it is still unclear the internal mechanism, Ru additions are expected to suppress the formation of TCP phase. The extensively accepted assumption is “reversed partitioning” method by which Ru can promote the partitioning of Re and other refractory

elements to γ' precipitates, thus reducing the tendency of segregation in the matrix. But some recent researches have shown that “reversed partitioning” only happens to some special conditions where a variety of alloying elements are included in alloys. In some quaternary or quinary alloy systems, the “reversed partitioning” was not obvious with additions of Ru. Thus it is more reasonable that “reversed partitioning” results from the synthetic effects of various alloying elements instead of the separated action of Ru. In recent, researches on new generations of Ni-SXs are mainly conducted on the optimization of the Re/Ru proportion without extra

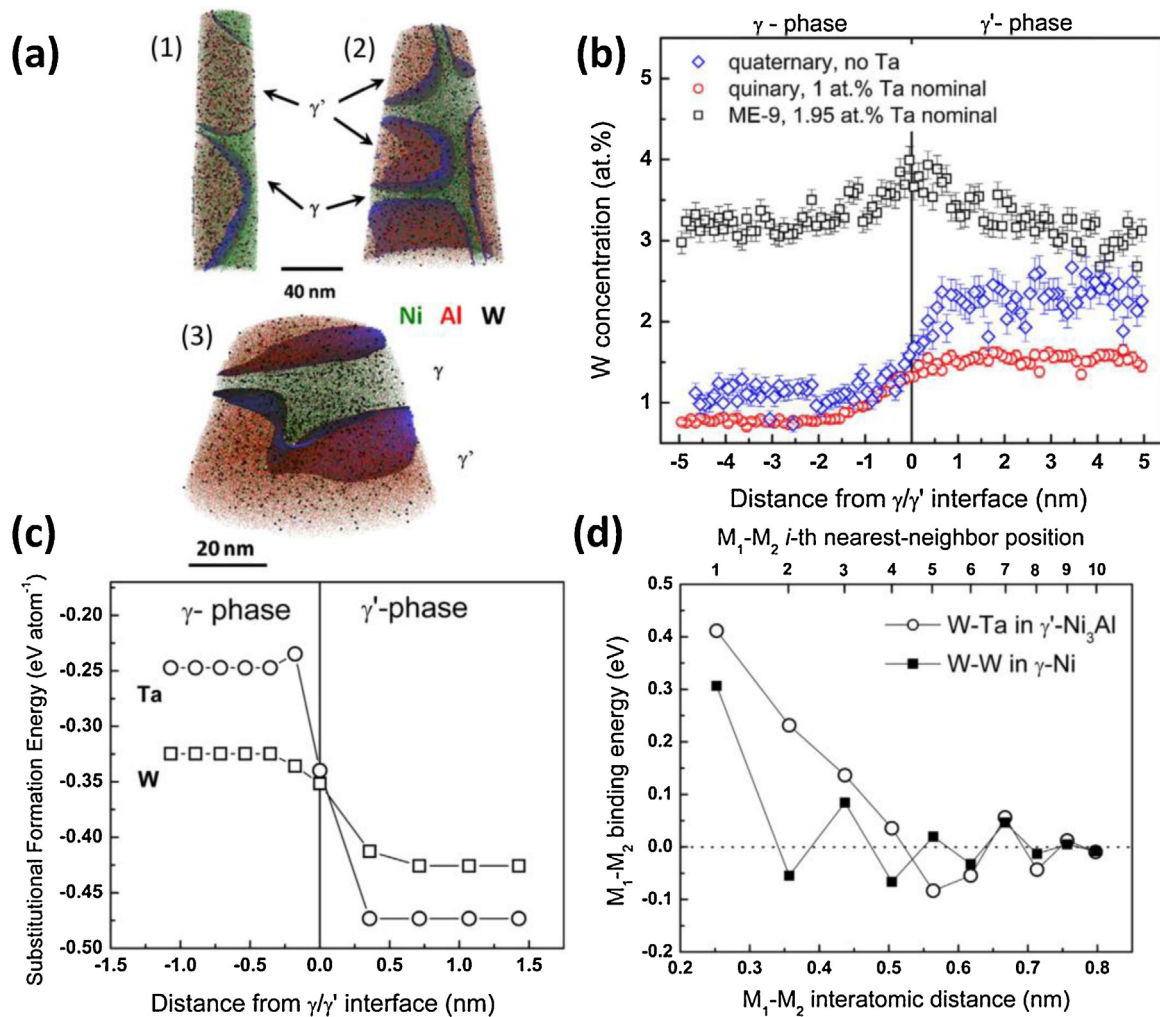


Fig. 21. Interactions between Ta and W distracted W atoms from γ' precipitates into γ matrix. Adapted from ref. [219]. (a) Projected 3D-APT reconstructions acquired from: (1) Ni-Al-Cr-W quaternary alloy; (2) Ni-Al-Cr-W-Ta quinary alloy (1 at.% Ta), both aged at 1073 K for 264 h; and (3) multicomponent ME-9 alloy (1.95 at.% Ta); (b) Proximity histograms of the W concentrations across the γ/γ' interfaces of the three types of alloy in (a); (c) The substitutional formation energies of W and Ta atoms as a function of their distances from the γ/γ' interface; (d) The binding energies of W-Ta dimers in the γ' phase and for W-W dimers in the γ phase as a function of interatomic distance (nm).

additions of these materials to control the cost and reduce the density of alloys.

It is very convenient to build the ternary or quaternary alloying systems for analyzing the effect of a single element on the microstructures and creep properties. But it should be noticed that the composition evolution in superalloys is always faced with very complex situations where many kinds of alloying elements work together to act on alloys instead of just three or four elements. One element which is added in different alloying systems might reversely affect the systems. And even the composition is unchanged in an alloying system, the increase and decrease of alloying contents would also give rise to different results. This is why the real effects of a single alloying element are difficult to be illustrated. The synthetic effects between different alloying elements should be highlight. This is because the inter-diffusion and the co-segregation of alloying elements directly influence the phase forming and microstructures that lattice misfit, the size and the fraction volume of γ' precipitate and the γ/γ' interface are expected to be changed. The resulted microstructures greatly influence the movements and interactions of dislocations and eventually determine the mechanisms of creep deformation. To consider the important role played by dislocations during creep of alloy, two primary methods in the elemental design of alloy should be highlighted. The first

one is to enhance the internal strength of the two phases that is to enhance the resistance to dislocation movements. In γ phase, elements such as Ru, Co and Re which can enhance the SFE effectively should be included. Also solutes such as Re which diffuse very slow in the γ phase help to restrict the diffusion of dislocations and play an important part in enhancing the creep resistance of alloy. To enhance the internal creep resistance of γ' phase, it is necessary to enhance the difficulty for dislocations moving into the precipitates. Alloying with Ta and/or Ti can effectively increase the APB energy that enhance the resistance of γ' phase to dislocations. Another method is to slow down the degradation of γ/γ' microstructure during creep, especially the rafting of γ' precipitates. Since rafting is promoted by accumulated plastic stains in γ channels, alloying additions which help to restrain the dislocation movements could be useful to retard the generation of plastic flows thus retarding the rafting process. Also, solutes such as Re which can stabilize the γ/γ' interface to retard the coarsening of γ' precipitates are expected to greatly enhance the creep strength of alloy. These aspects need to be synthetically considered in alloy design and it is very valuable to analyze the synthetic effects between two or more alloying elements on the creep properties. Future researches need to work on three aspects to discuss the creep properties in Ni-SXs. At first, the analysis of elemental segregation behavior in γ matrix and γ' pre-

cipitates is important, since the distribution of elements now can be clearly characterized in alloys. Besides, the creep deformation of alloys is in nature related to the degradation of γ/γ' microstructure such as the coarsening and dissolution of the γ' precipitates. Thus it is useful to build accurate models of degradation according to the coarsening kinetics of γ' phase, the lattice misfit, and the dislocation motion as well as formation of dislocation networks. At last, it is significant to determine the creep deformation mechanism in alloys with concerning the elemental synthetic effects and the degradation of microstructures and eventually to take advantages of these relationships to guide the composition evolution in superalloy design.

Acknowledgements

This work was jointly supported by the Fundamental Research Funds for the Central Universities (No. 2019QNA4012) and the Innovation Fund of the Zhejiang Kechuang New Materials Research Institute (No. ZKN-18-Z01).

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